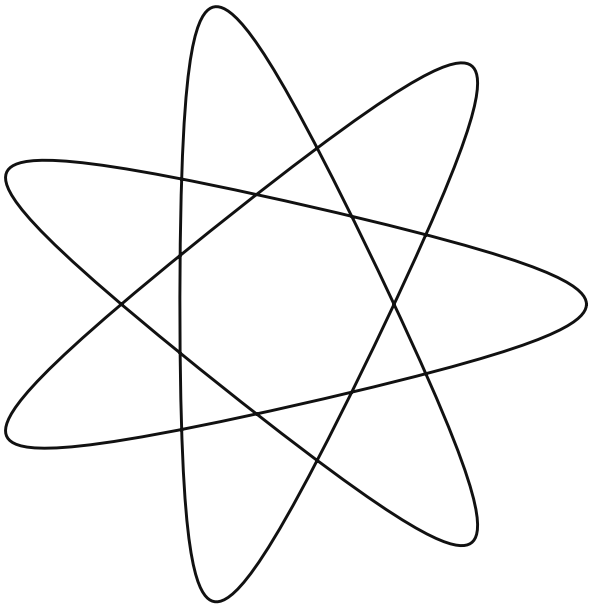
MATHEMATICAL PATTERNS

A Coloring Book of 50 Geometric Designs

fractals · spirals · tilings · mandalas



Junyi Zhou

MATHEMATICAL PATTERNS — A Coloring Book of 50 Geometric Designs

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Every pattern in this book is generated from the mathematical equations named on its page.

How to Use This Book

Each design sits on a right-hand page with a blank back,
so markers will not bleed through onto the next pattern.

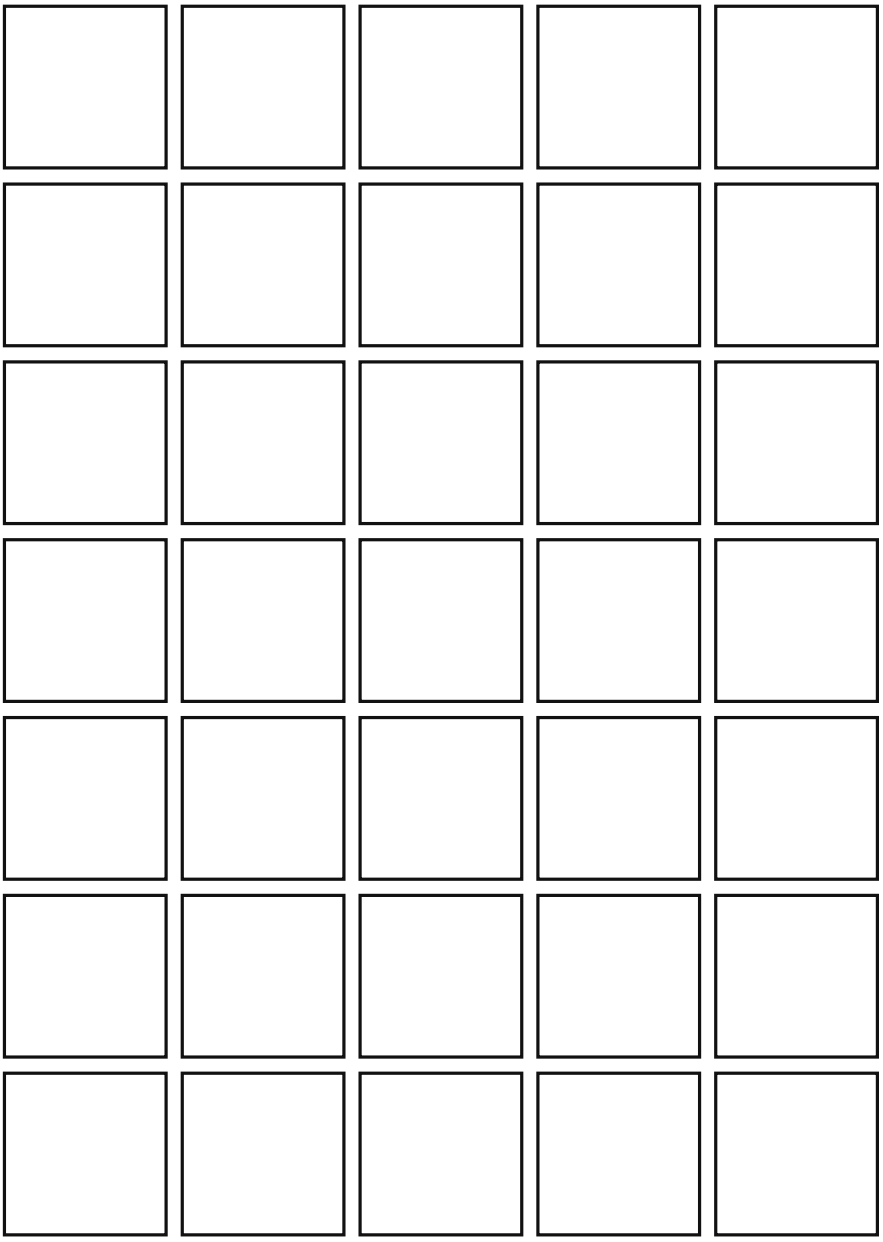
Below every design is the mathematical idea that drew it —
a one-line answer to “what am I coloring?”

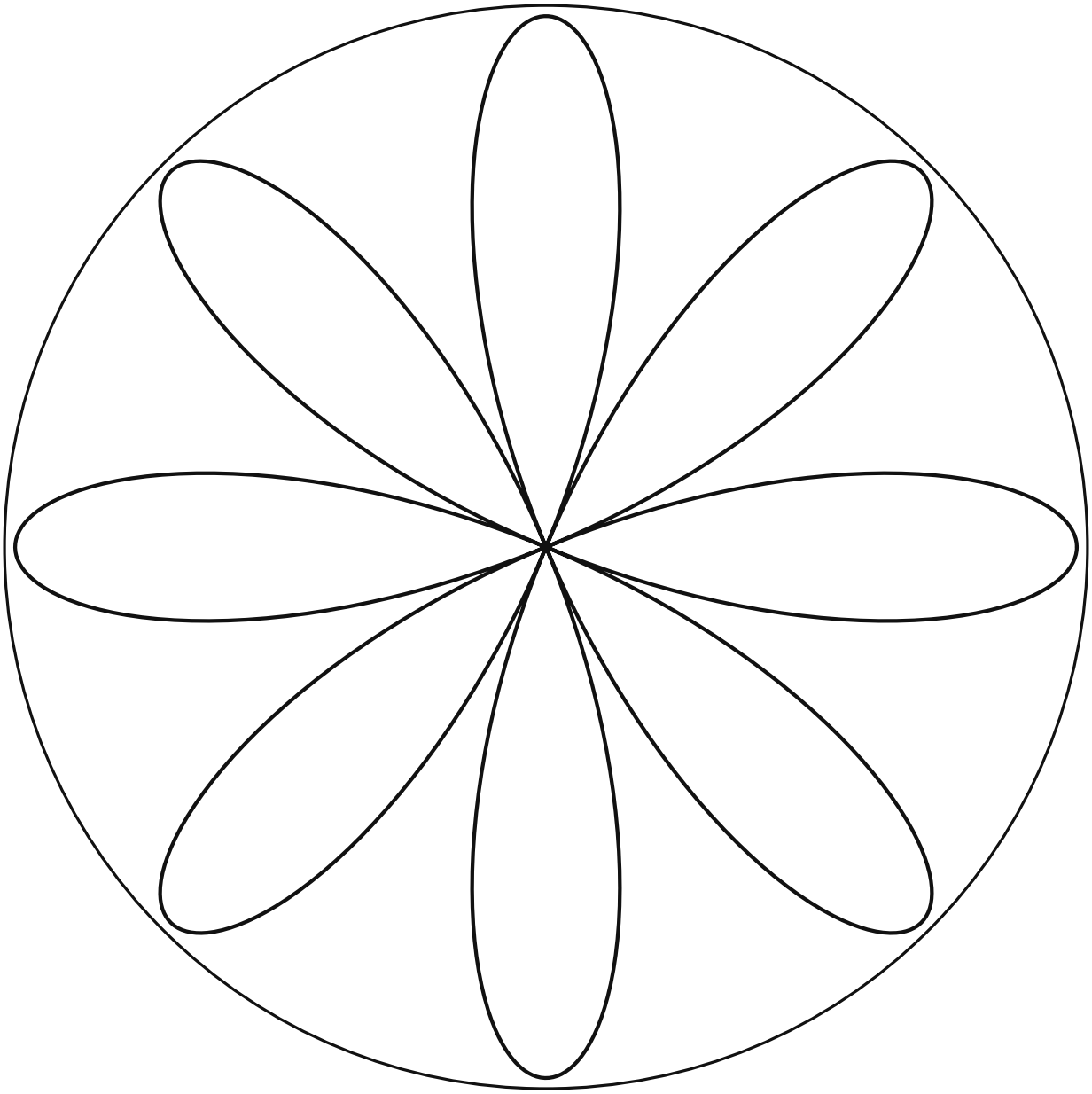
Try the color-test page opposite before using new pens.

There is no wrong way to color mathematics.

Color Test Page

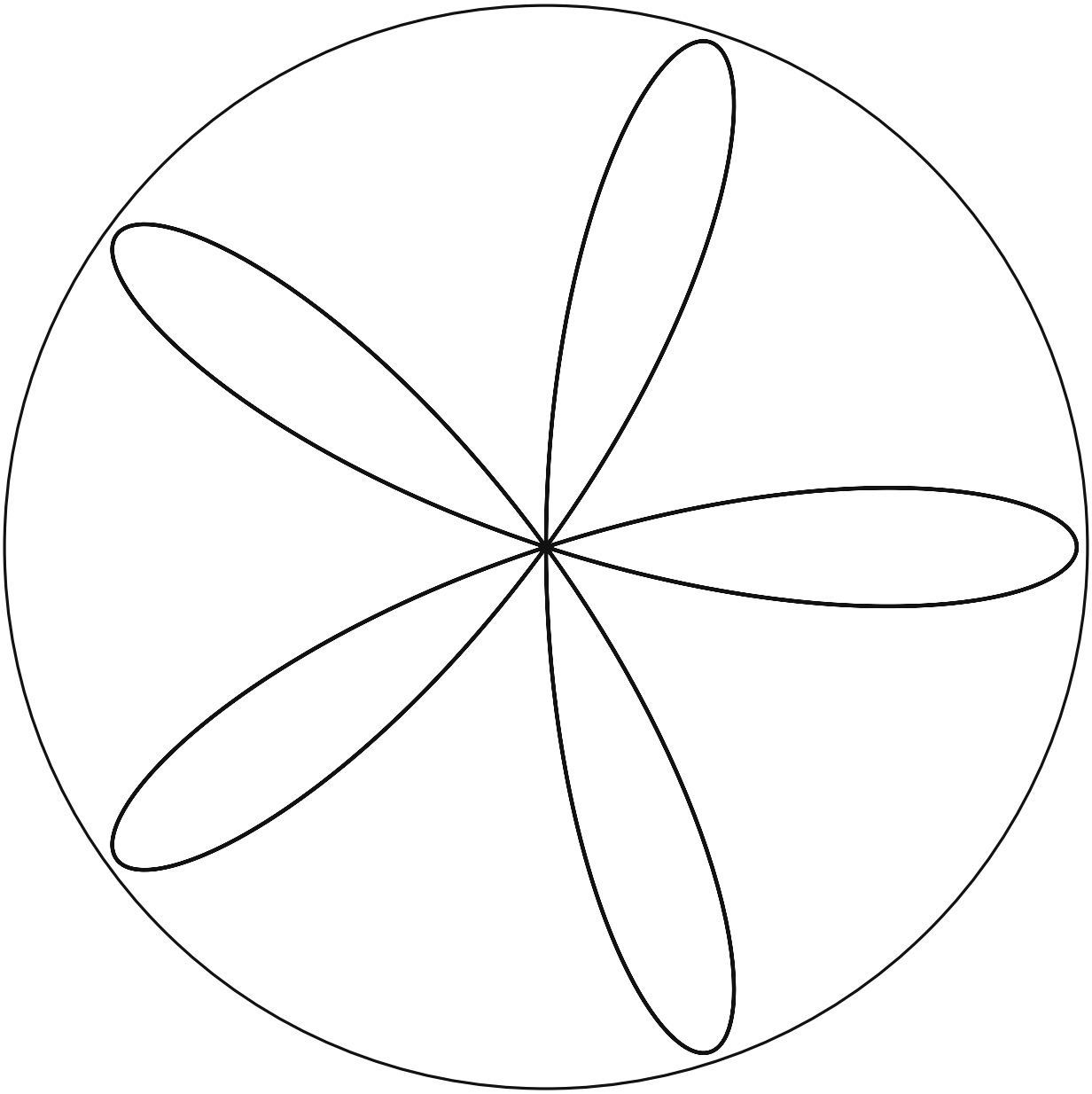
Try your pens and markers here first.





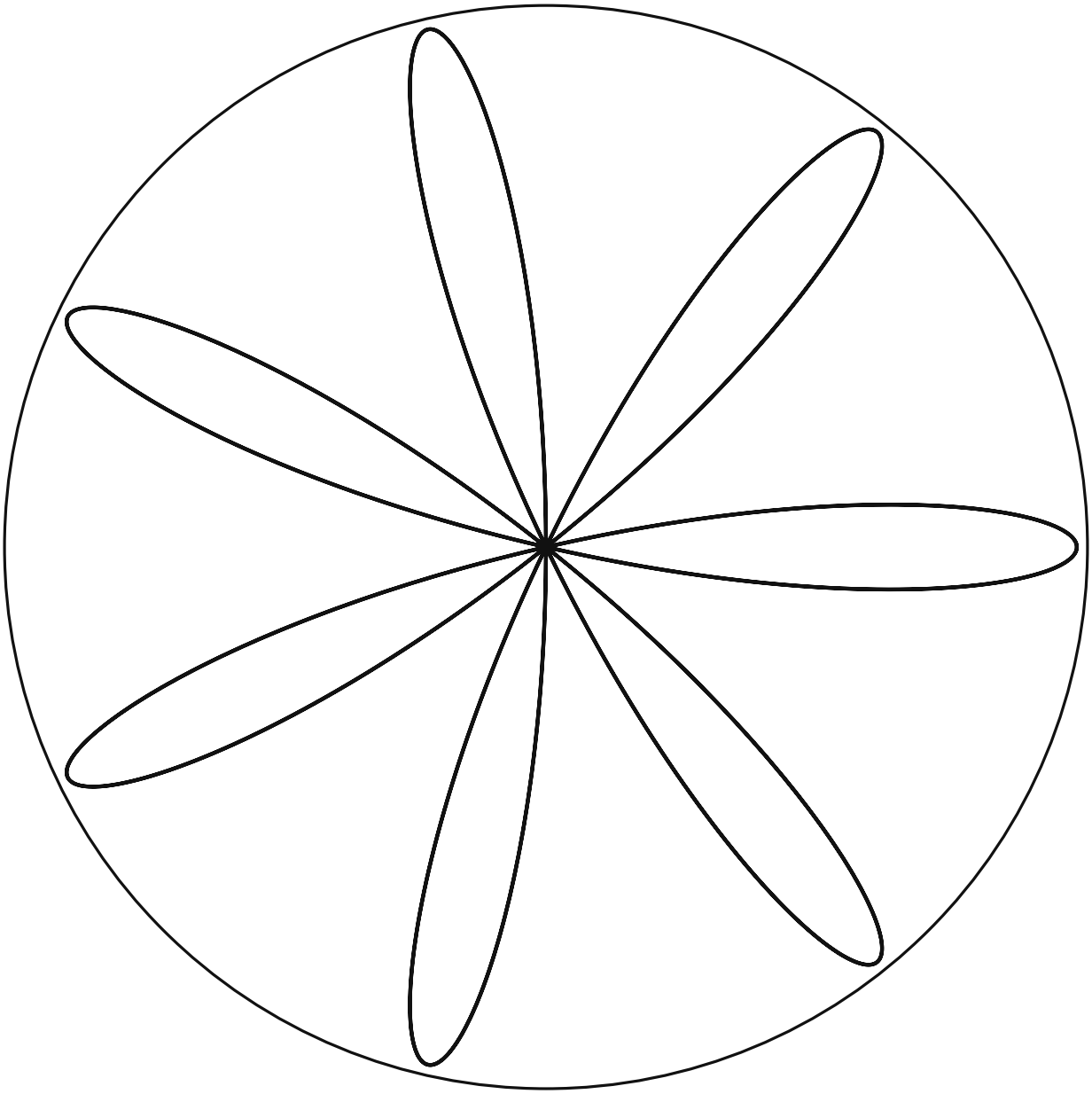
Rose Curve, 8 Petals

r = cos(4θ) — an even k doubles the petal count.



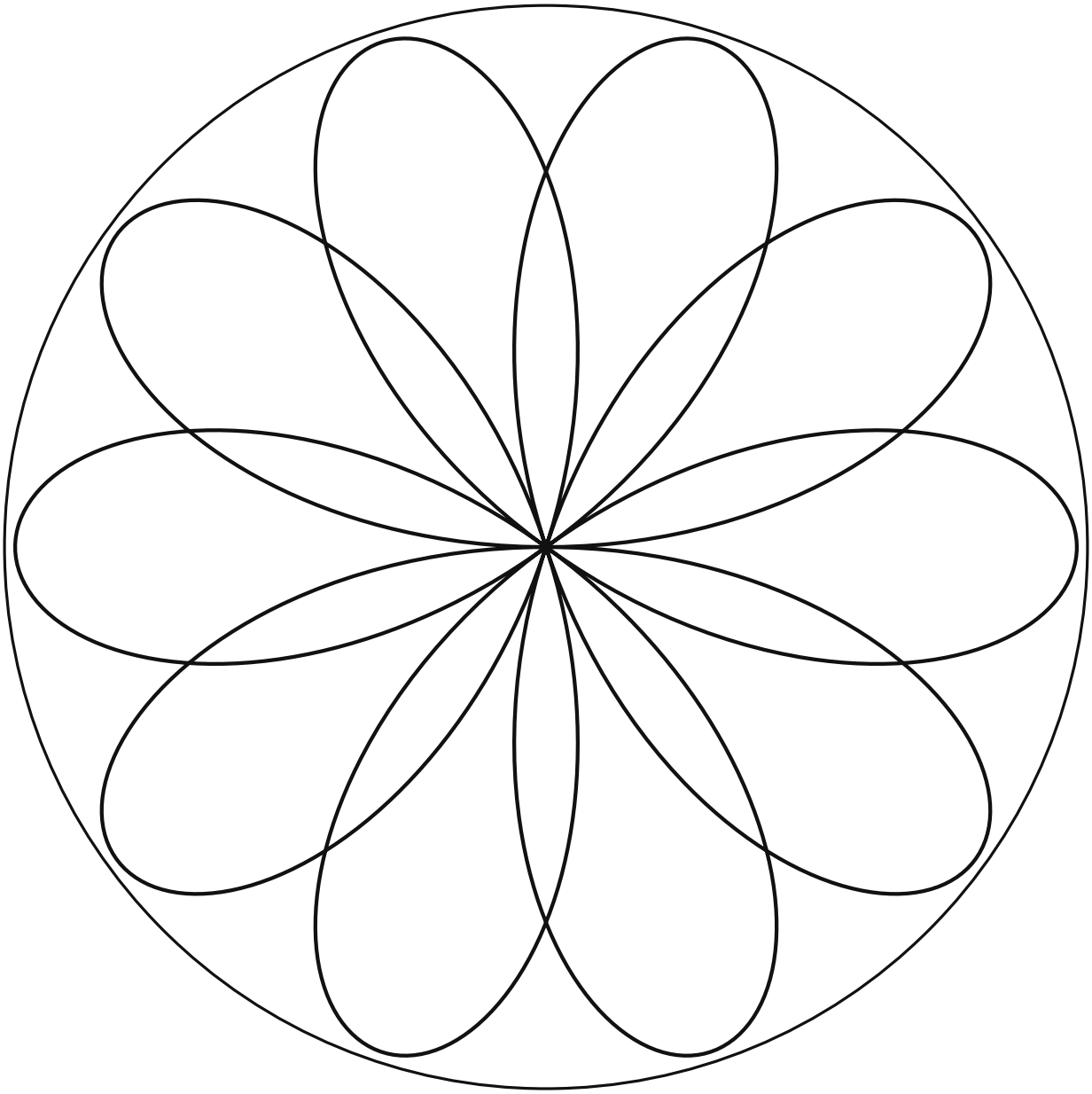
Rose Curve, 5 Petals

r = cos(5θ) — an odd k keeps exactly k petals.



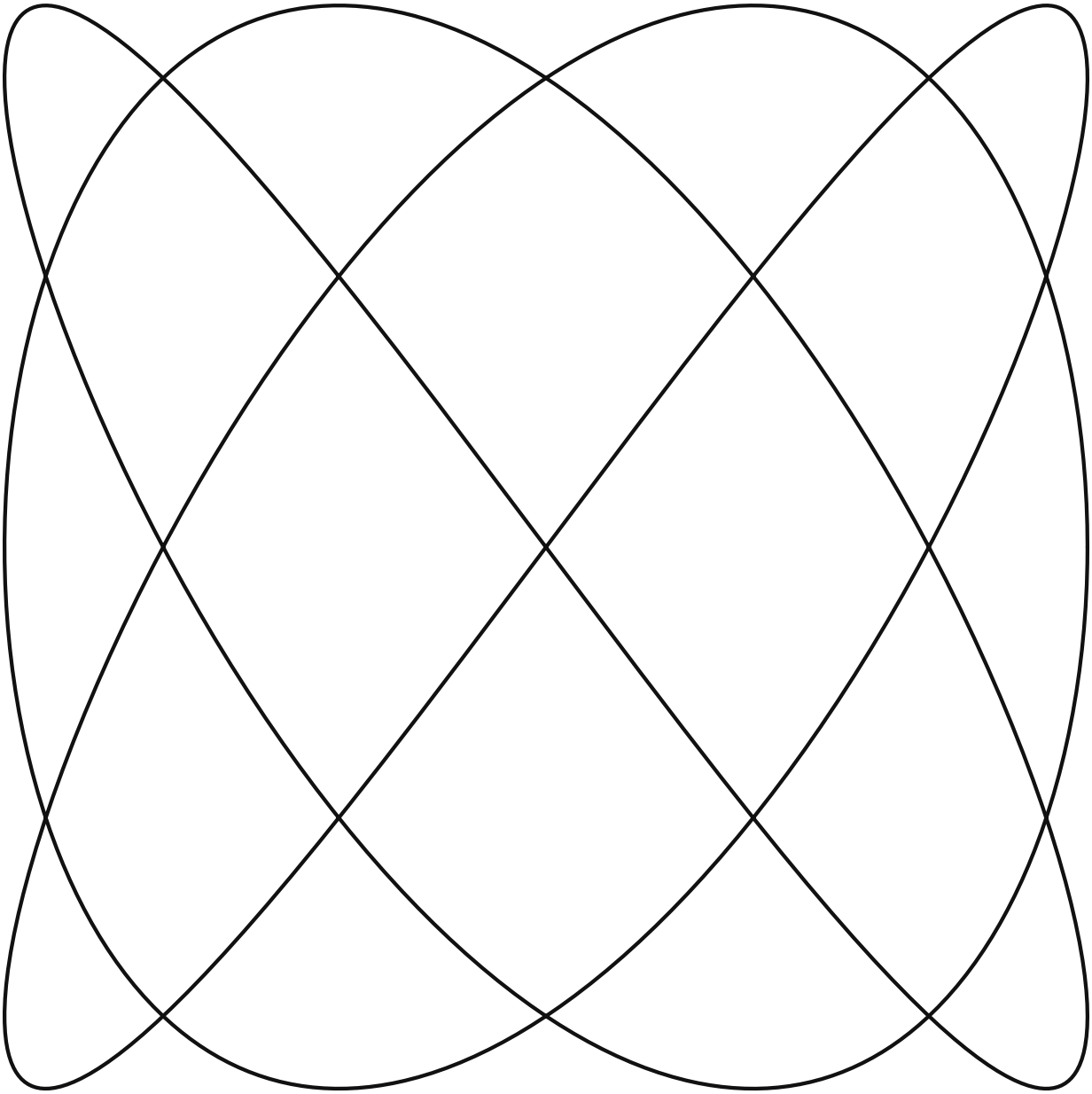
Rose Curve, 7 Petals

r = cos(7θ) — polar graphs turn equations into flowers.



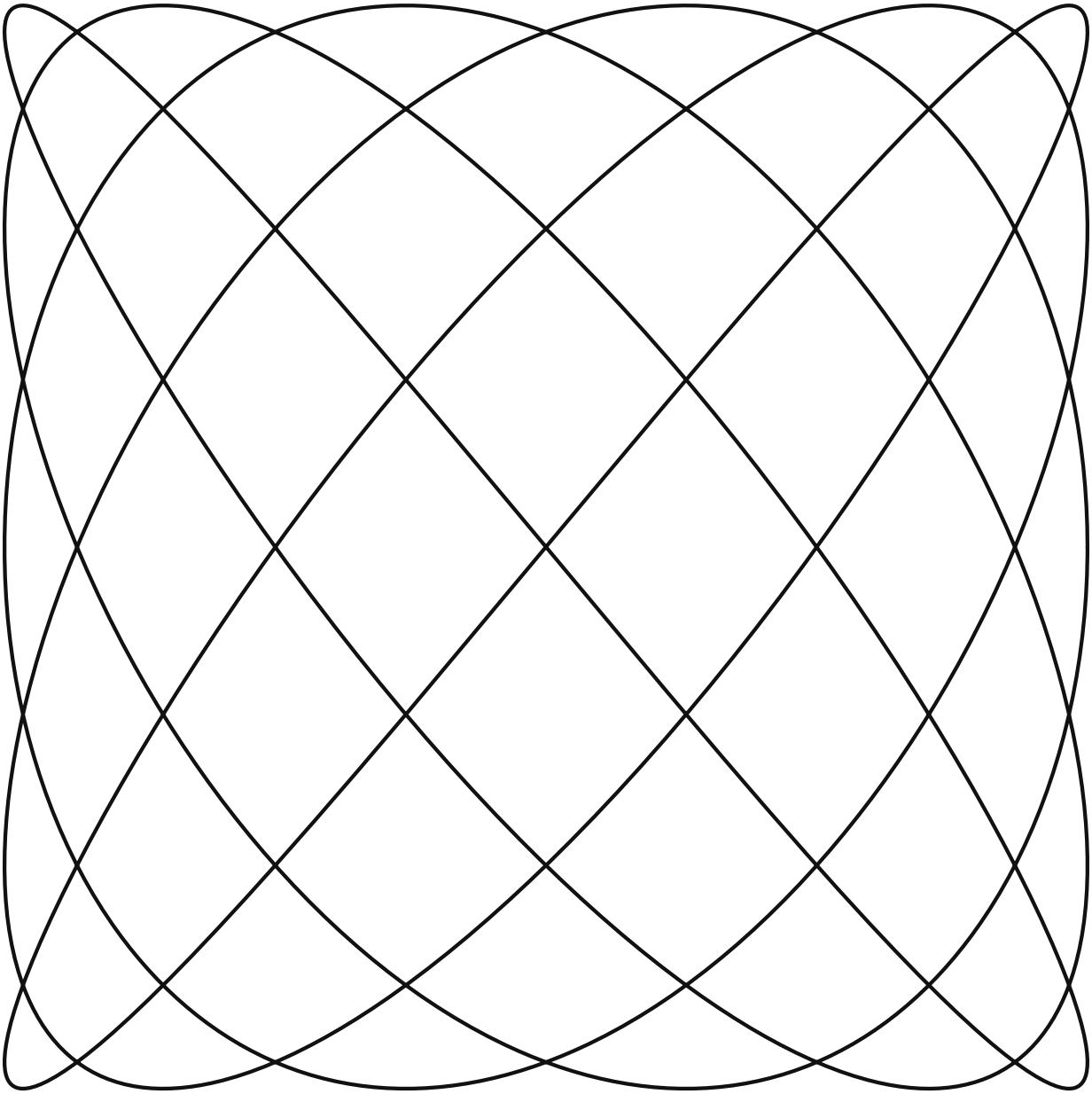
Overlapping Rose

r = cos(5θ/2) — a fractional k needs two turns to close.



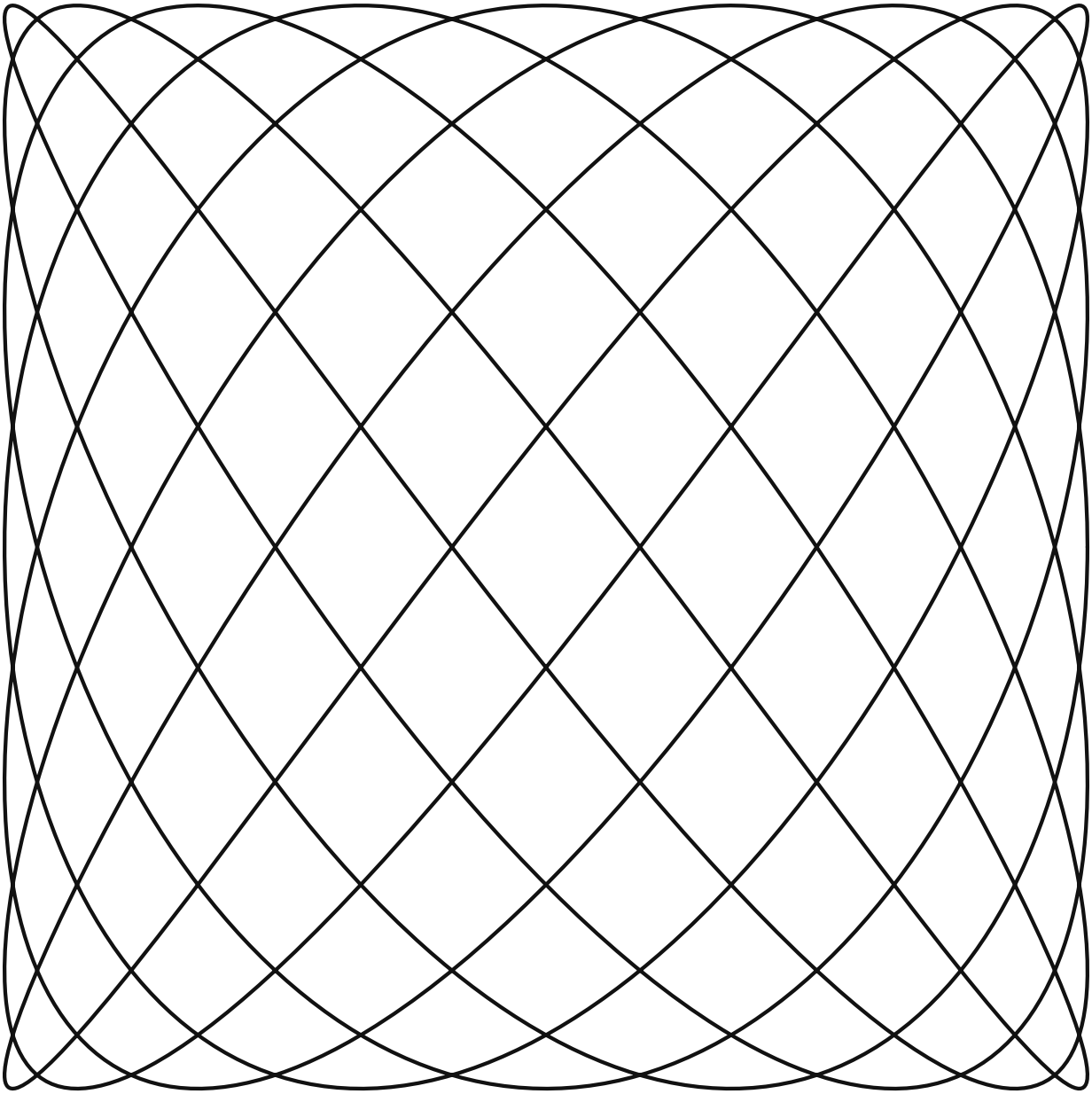
Lissajous 3:4

x = sin 3t, y = sin 4t — two perpendicular vibrations.



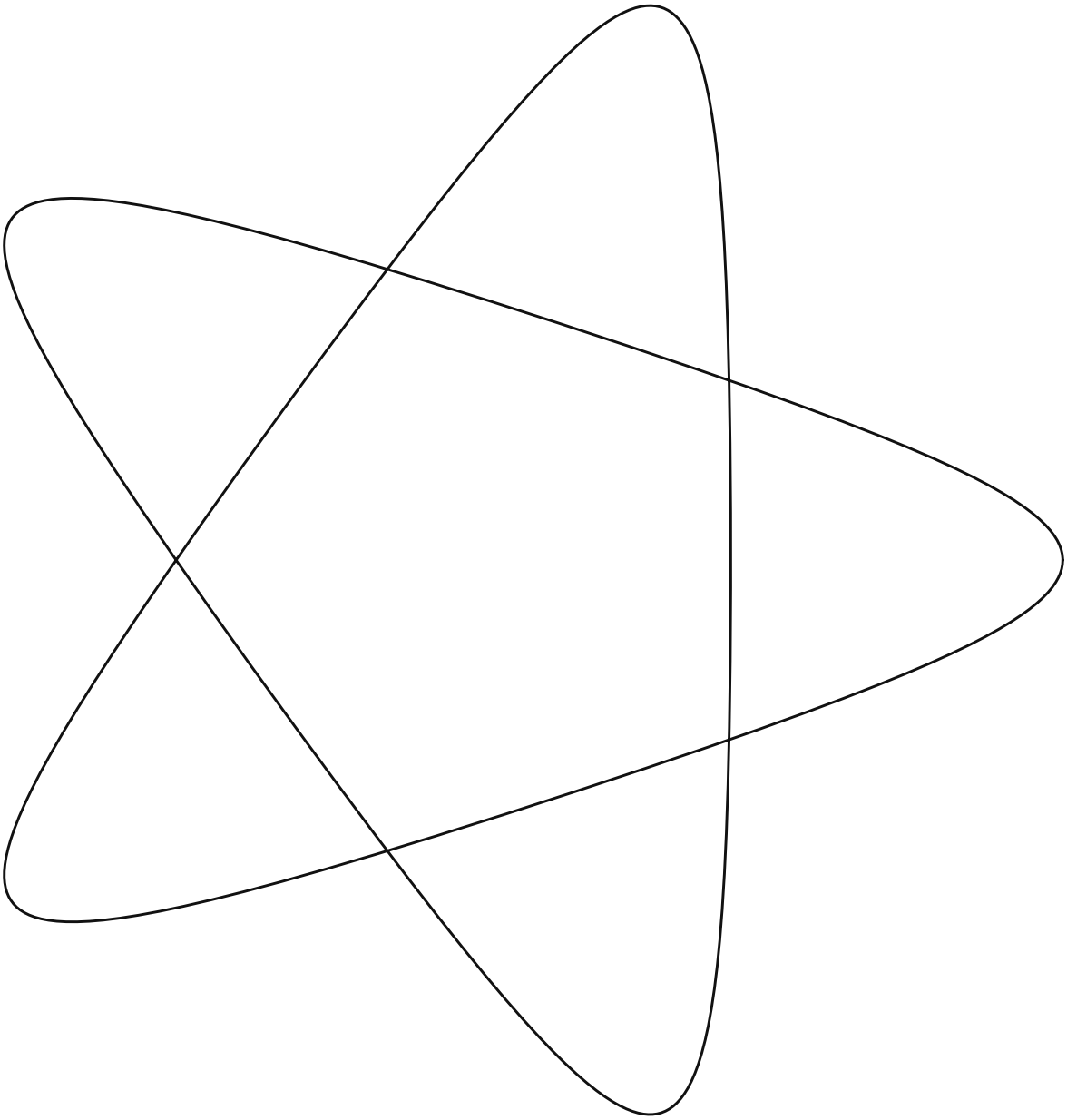
Lissajous 5:6

Neighboring frequencies weave a tight braid.



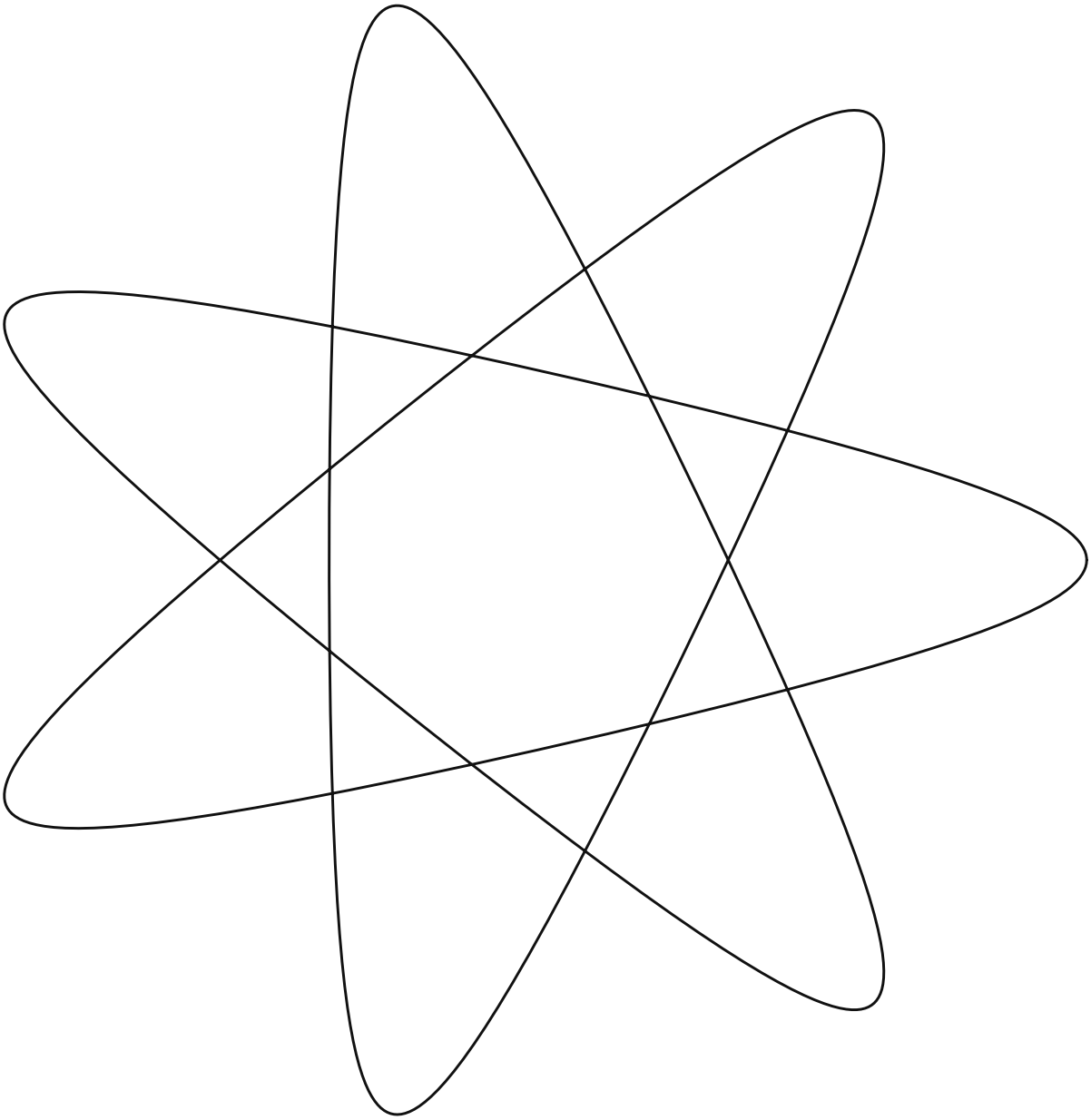
Lissajous 7:9

The pattern closes because 7 and 9 share no factor.



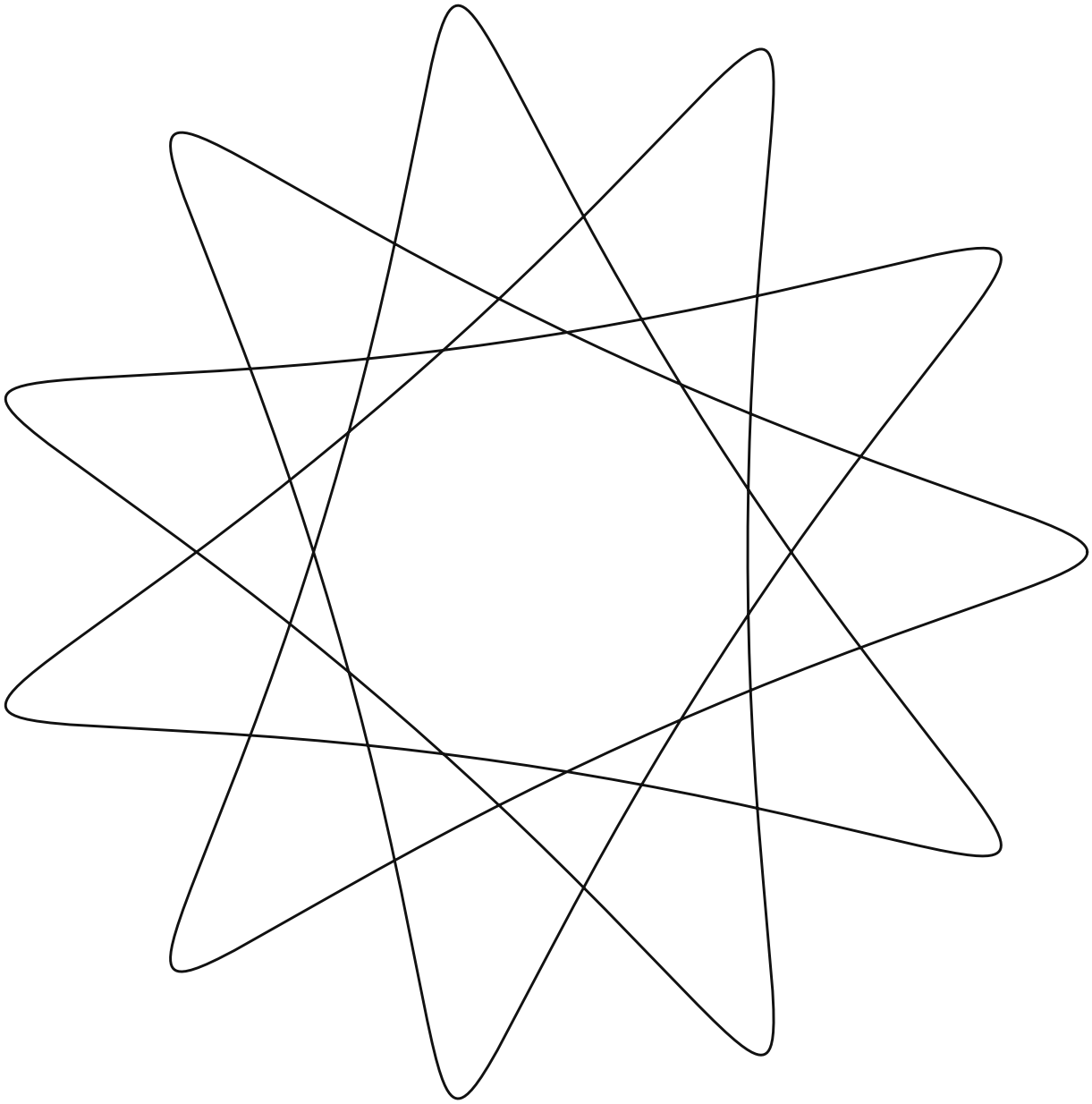
Spirograph 5:3

A circle rolling inside a circle traces a hypotrochoid.



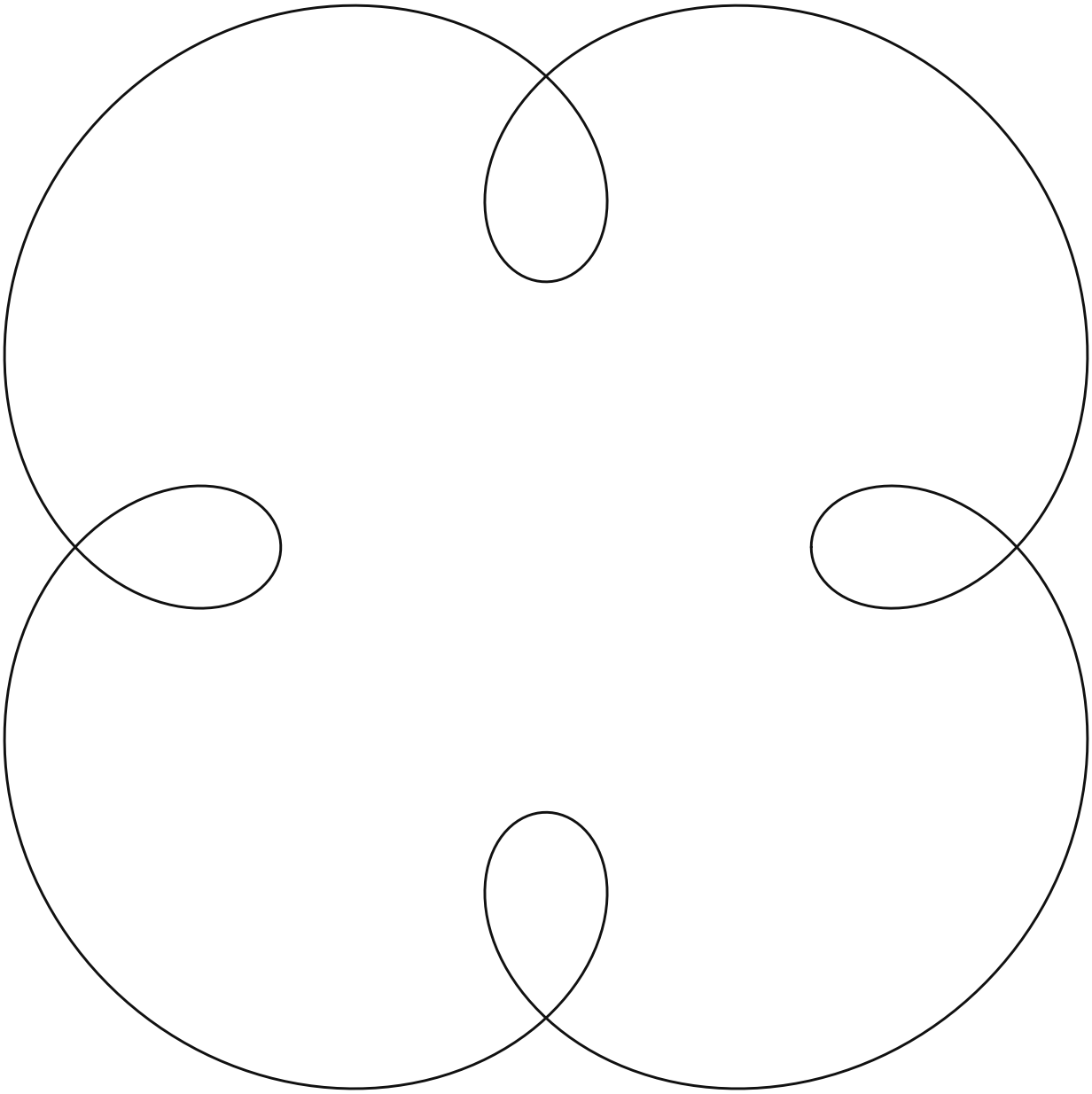
Spirograph 7:4

Change one gear tooth and the whole flower changes.



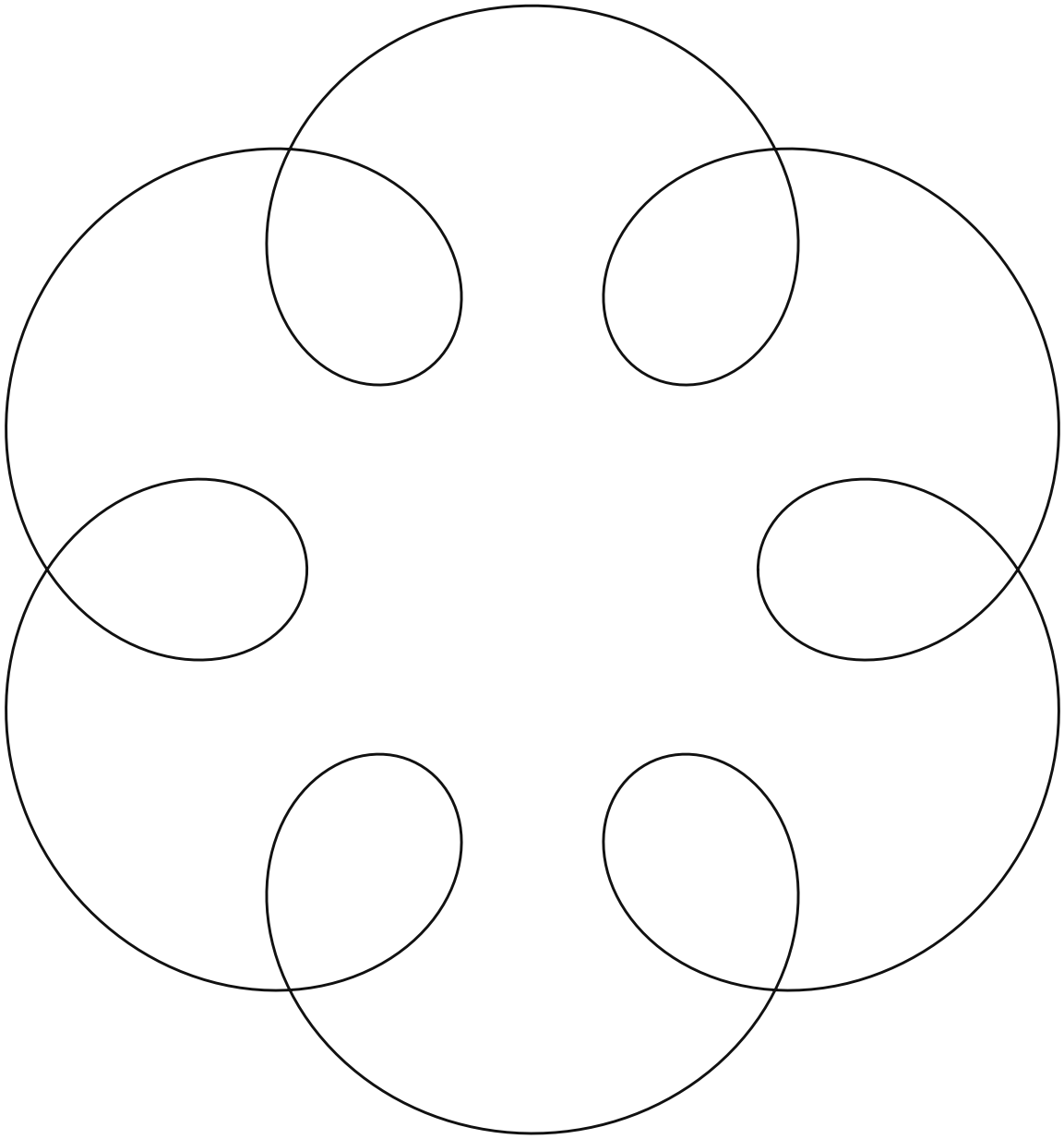
Spirograph 11:7

Coprime gears visit every position before closing.



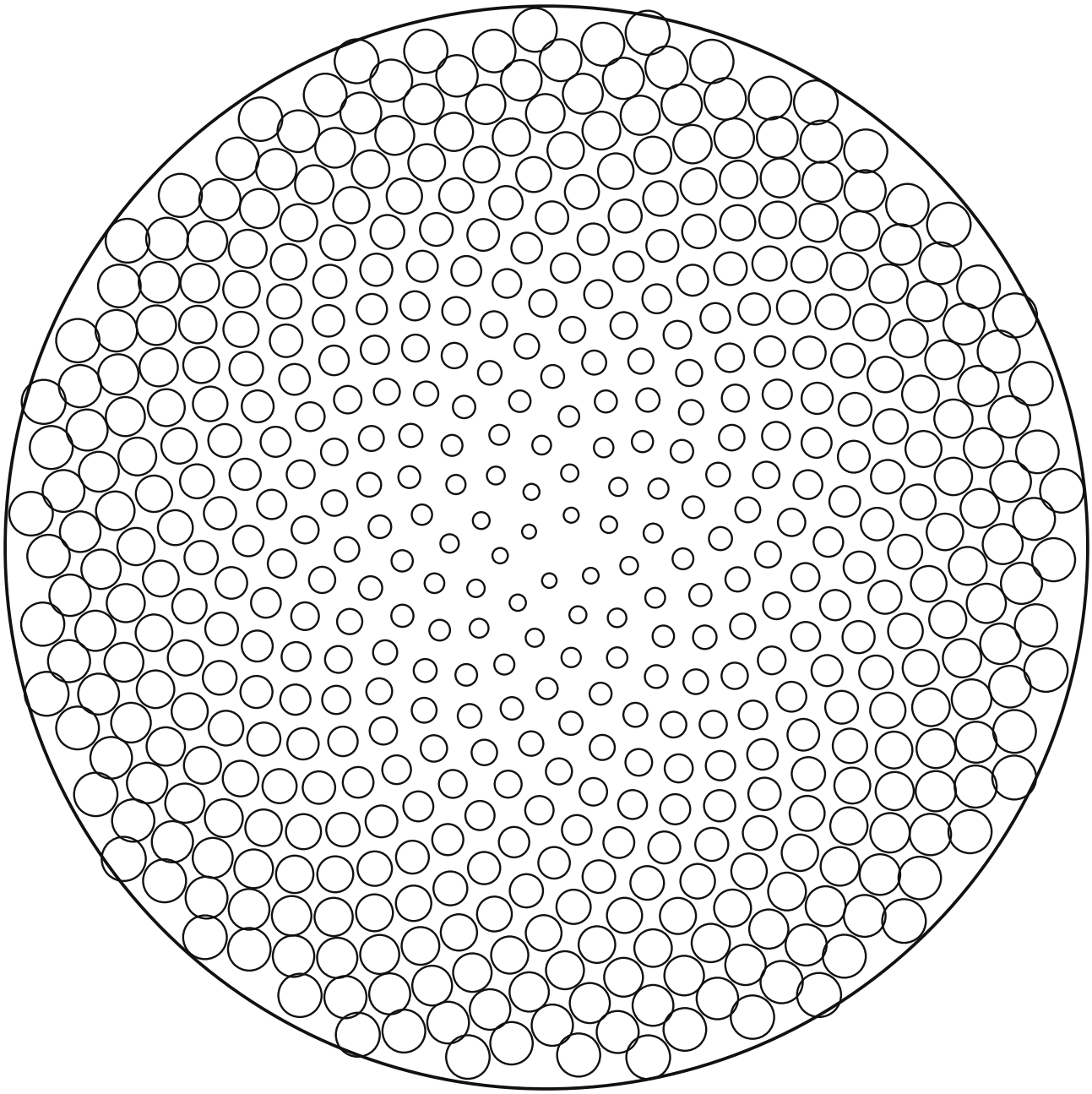
Epitrochoid 4:1

Here the small circle rolls outside the big one.



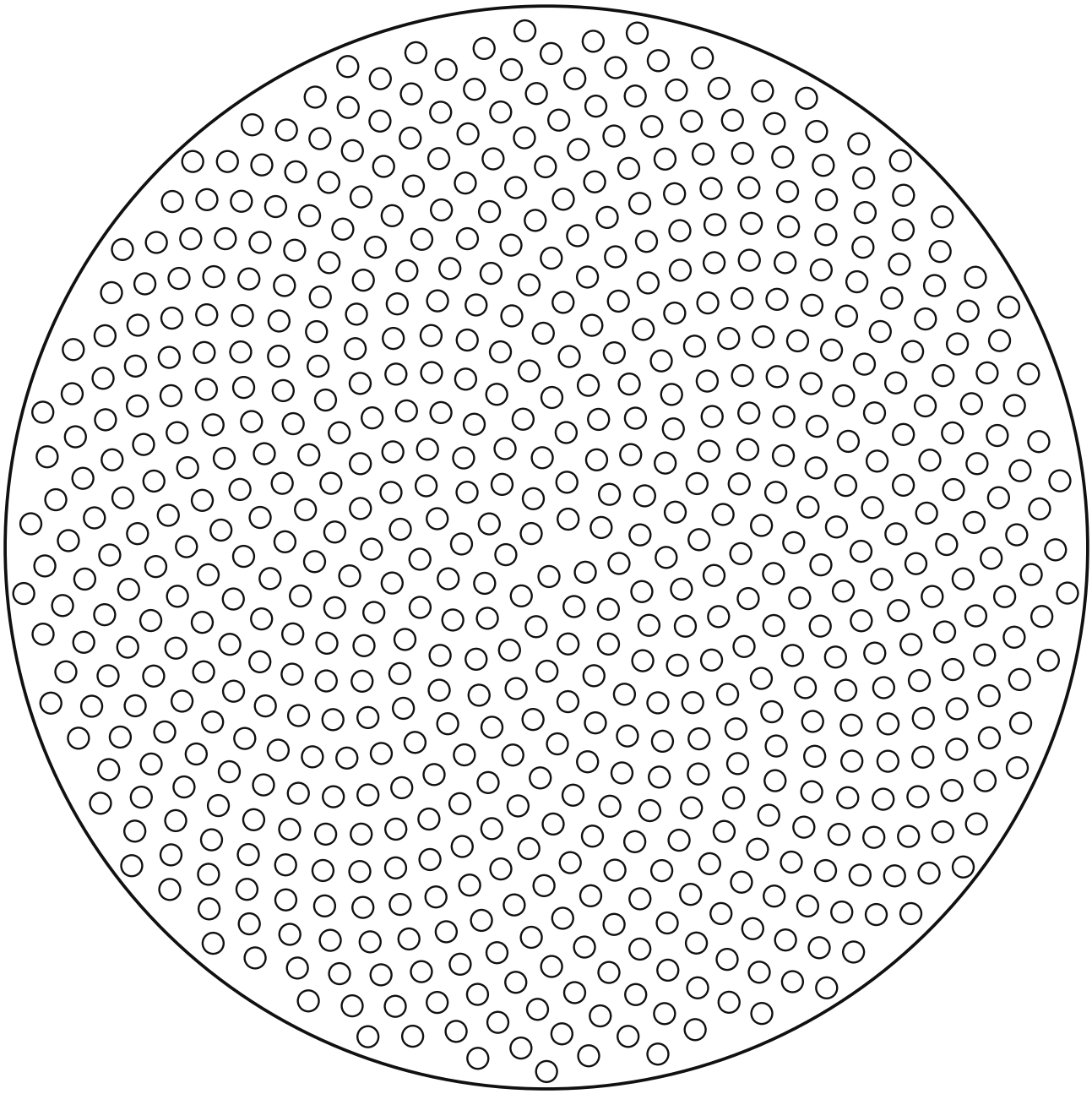
Epitrochoid 6:1

Six lobes: the ratio of the radii decides the count.



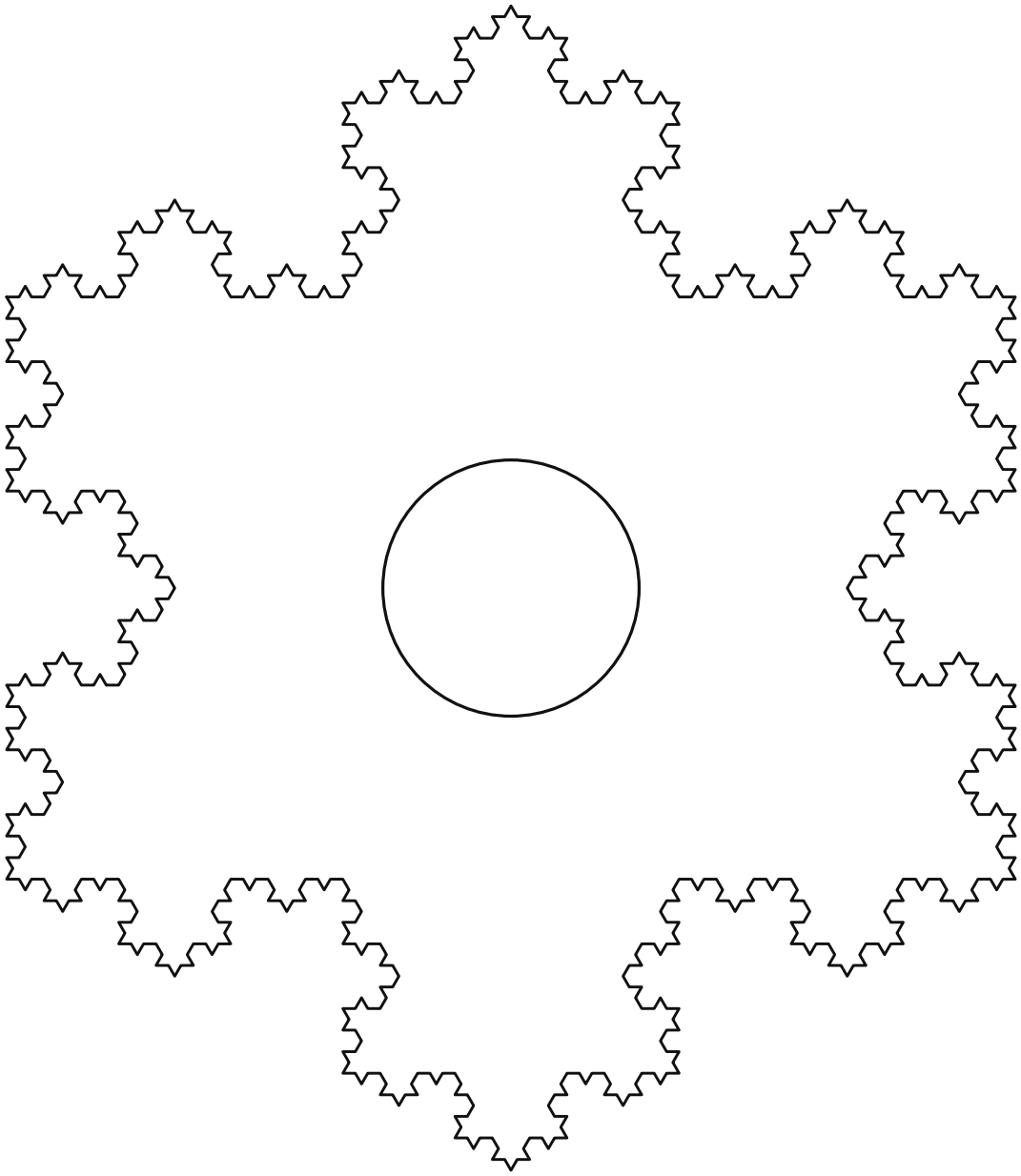
Phyllotaxis

Sunflower seeds: each turns 137.5° — the golden angle.



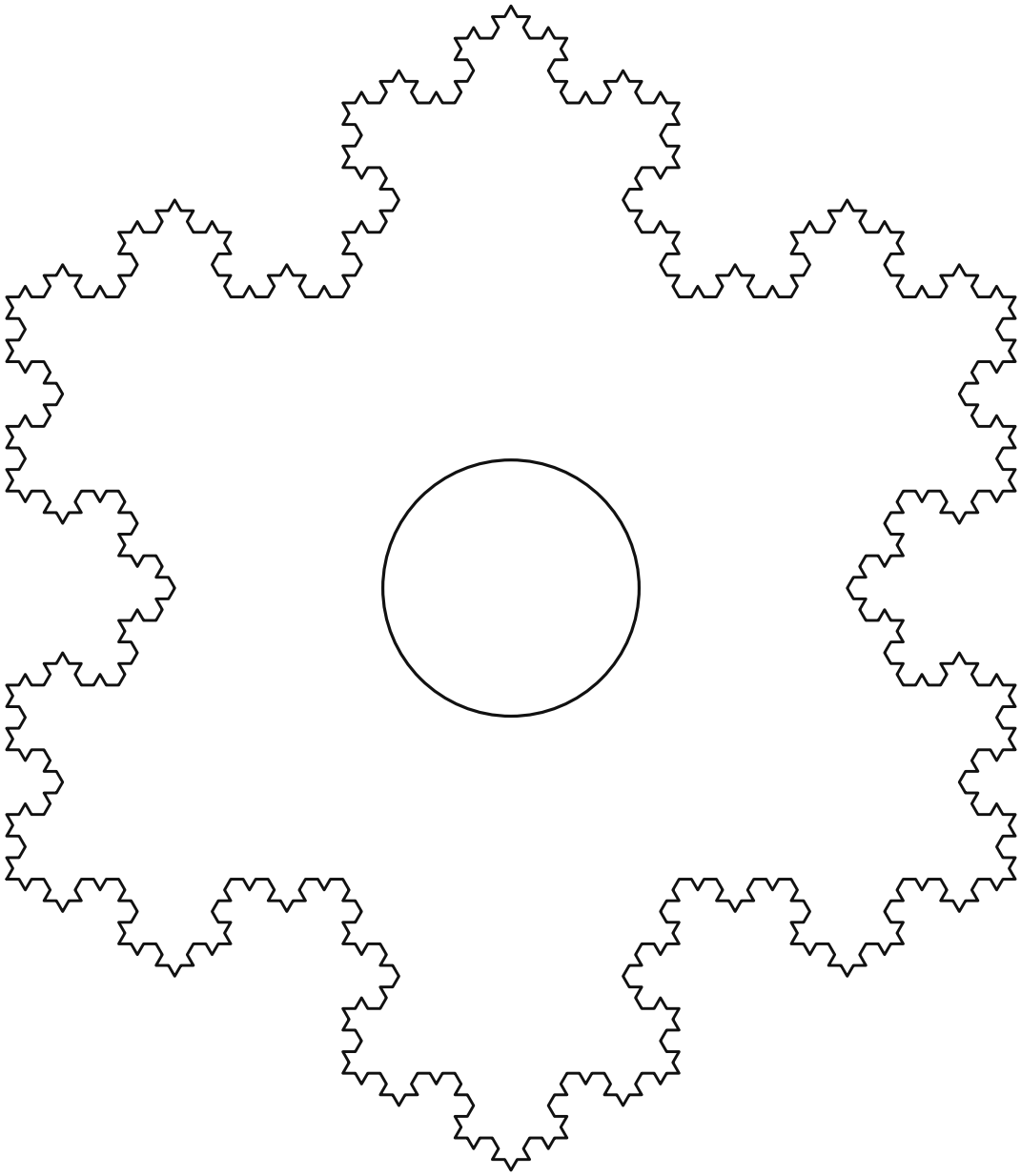
Phyllotaxis, Even Seeds

The same spiral with equal seeds shows the lattice.



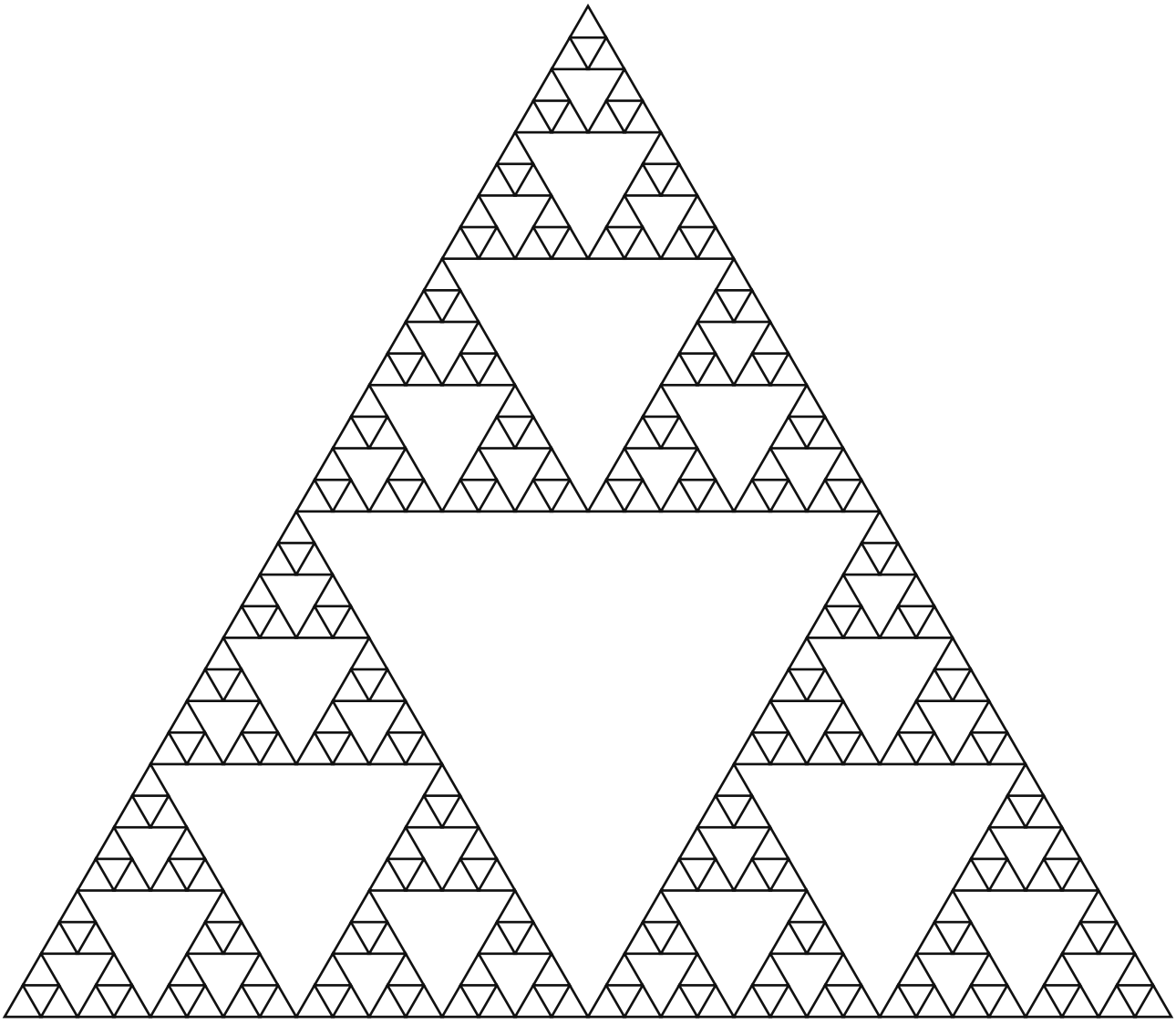
Koch Snowflake

Every edge sprouts a smaller copy of itself, forever.



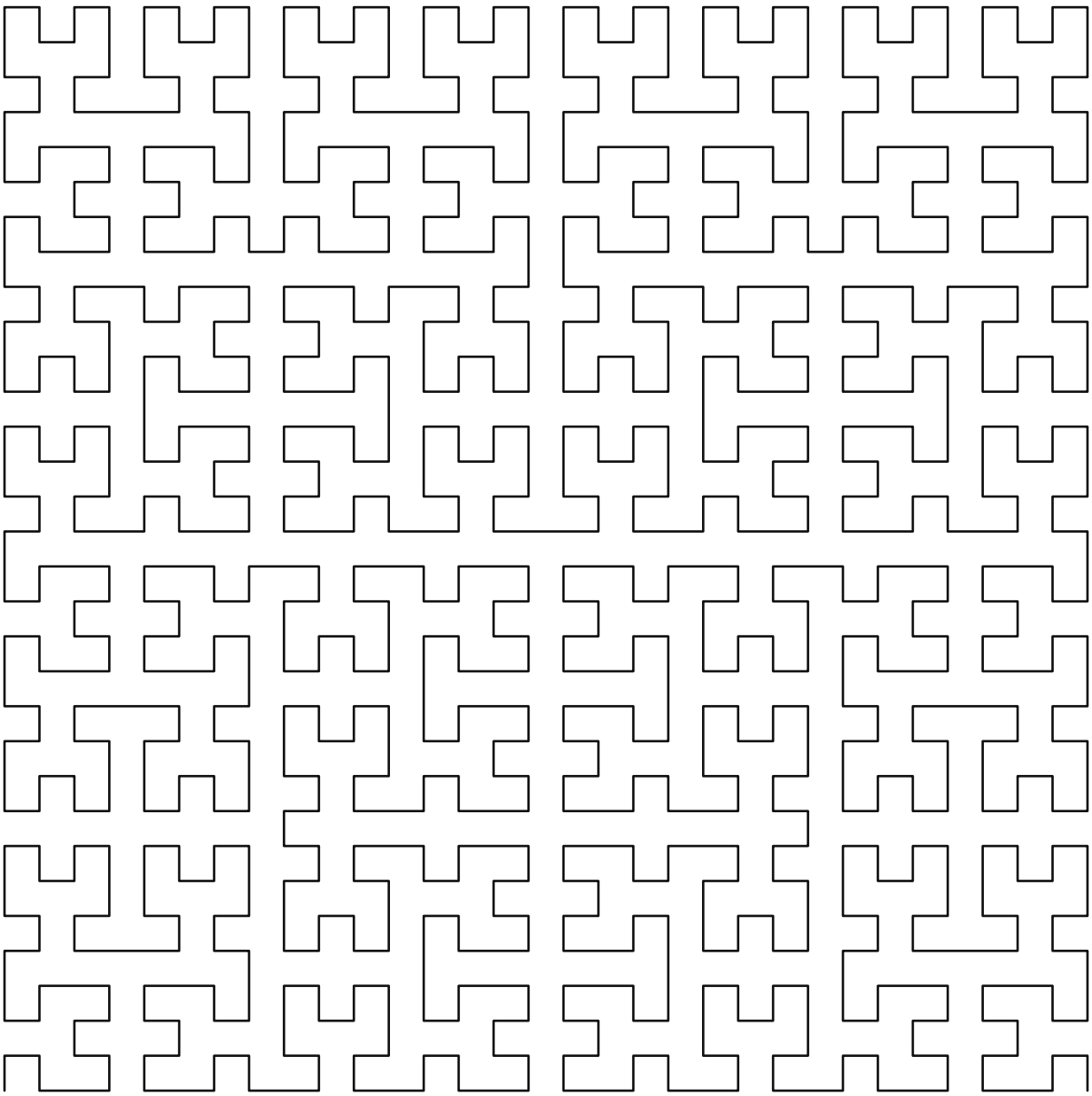
Koch Anti-Snowflake

The same rule bent inward eats the triangle instead.



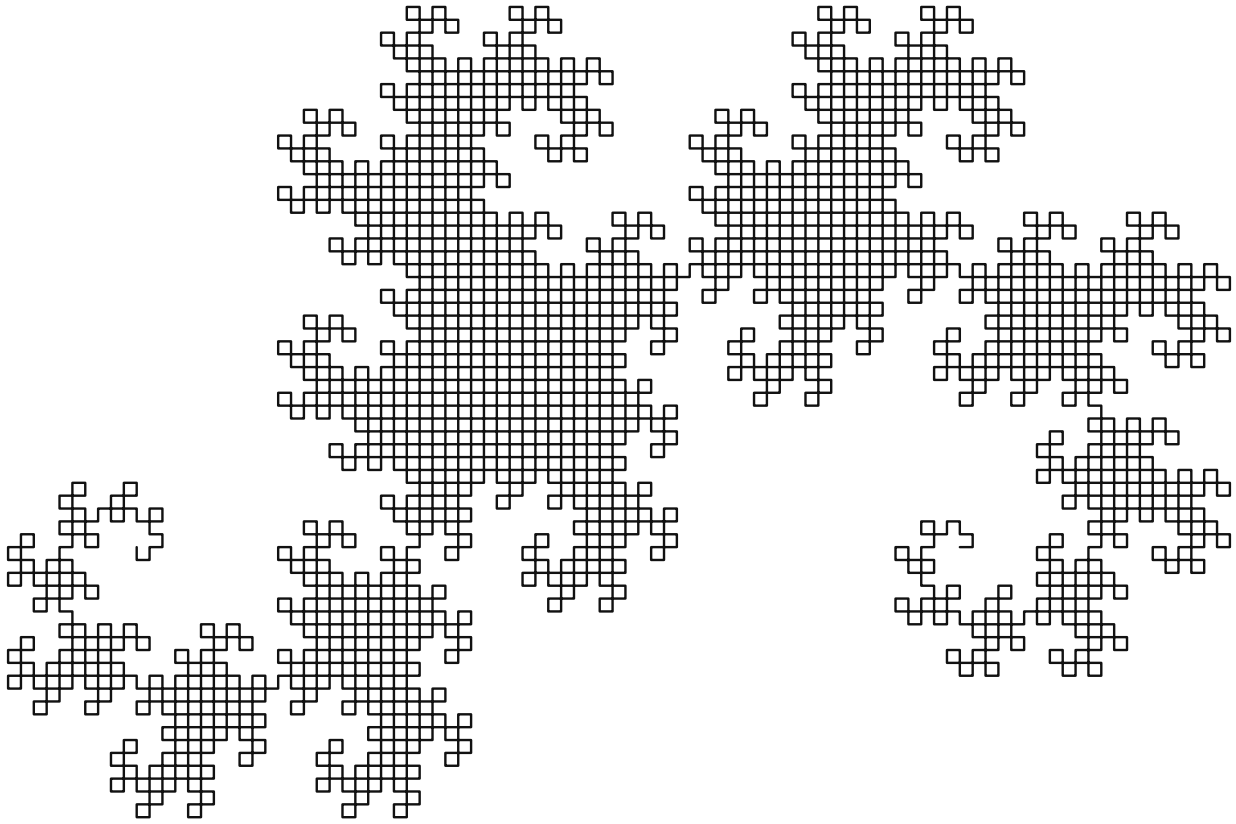
Sierpinski Triangle

Remove the middle triangle; repeat in each corner.



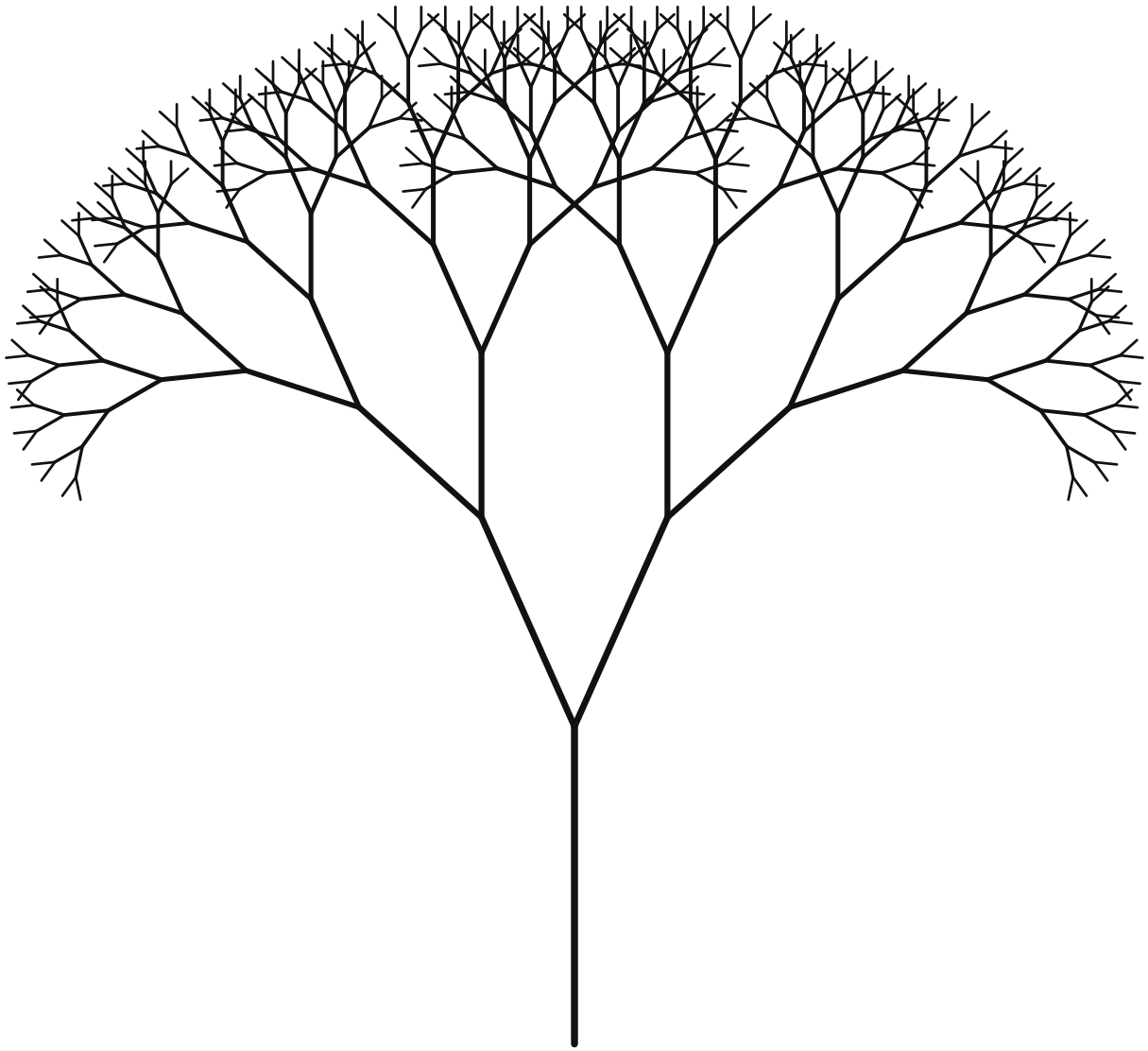
Hilbert Curve

One line that visits every cell of the square.



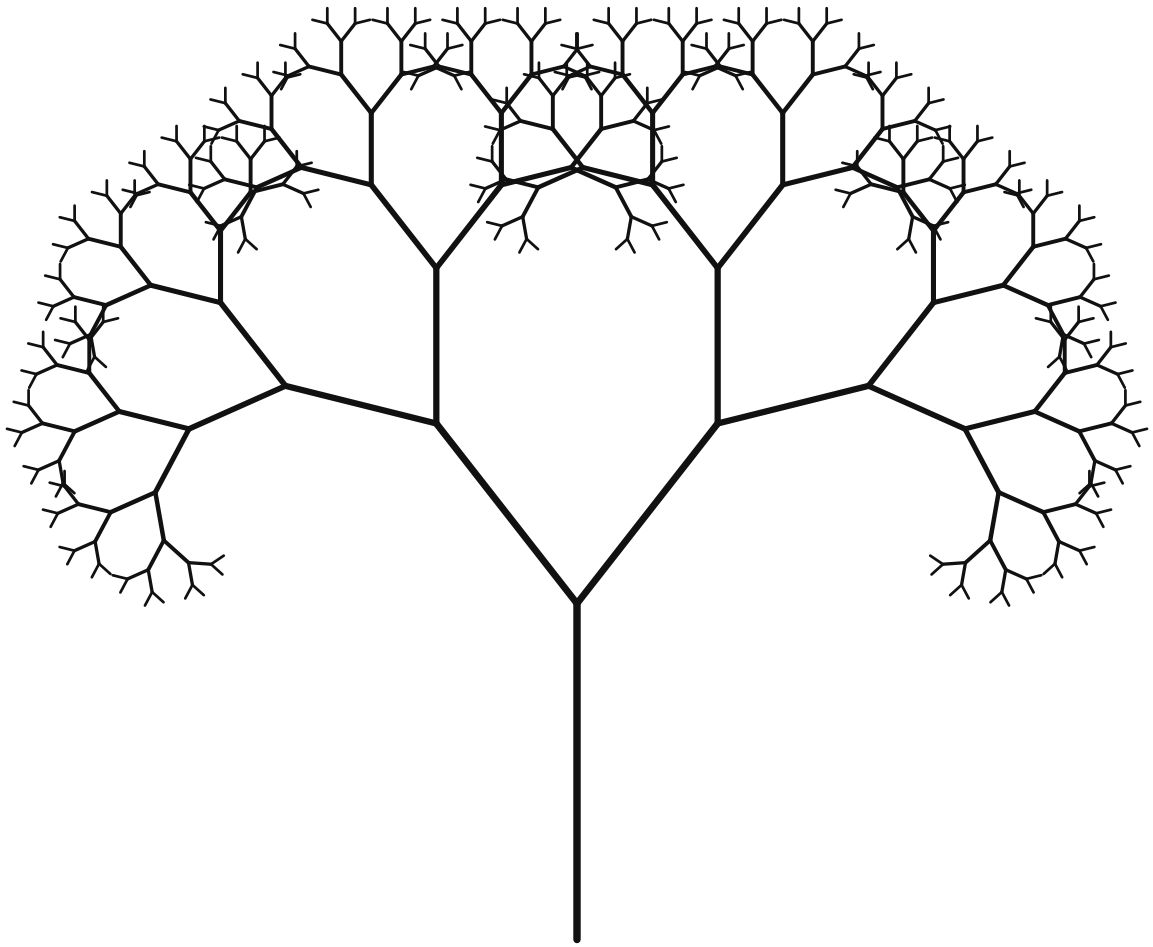
Dragon Curve

Fold a paper strip 12 times — this is its crease pattern.



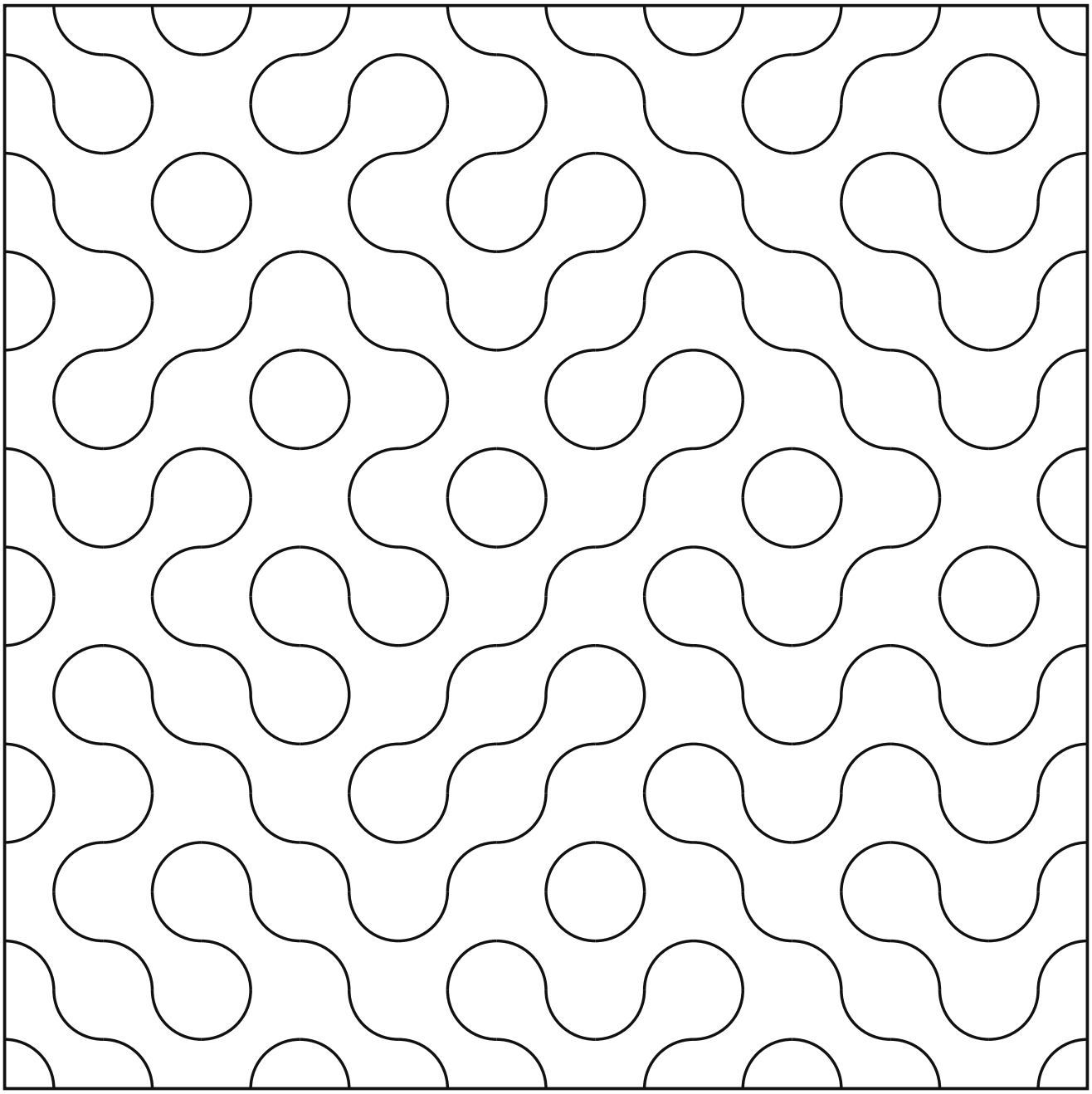
Fractal Tree

Each branch splits into two at the same angle.



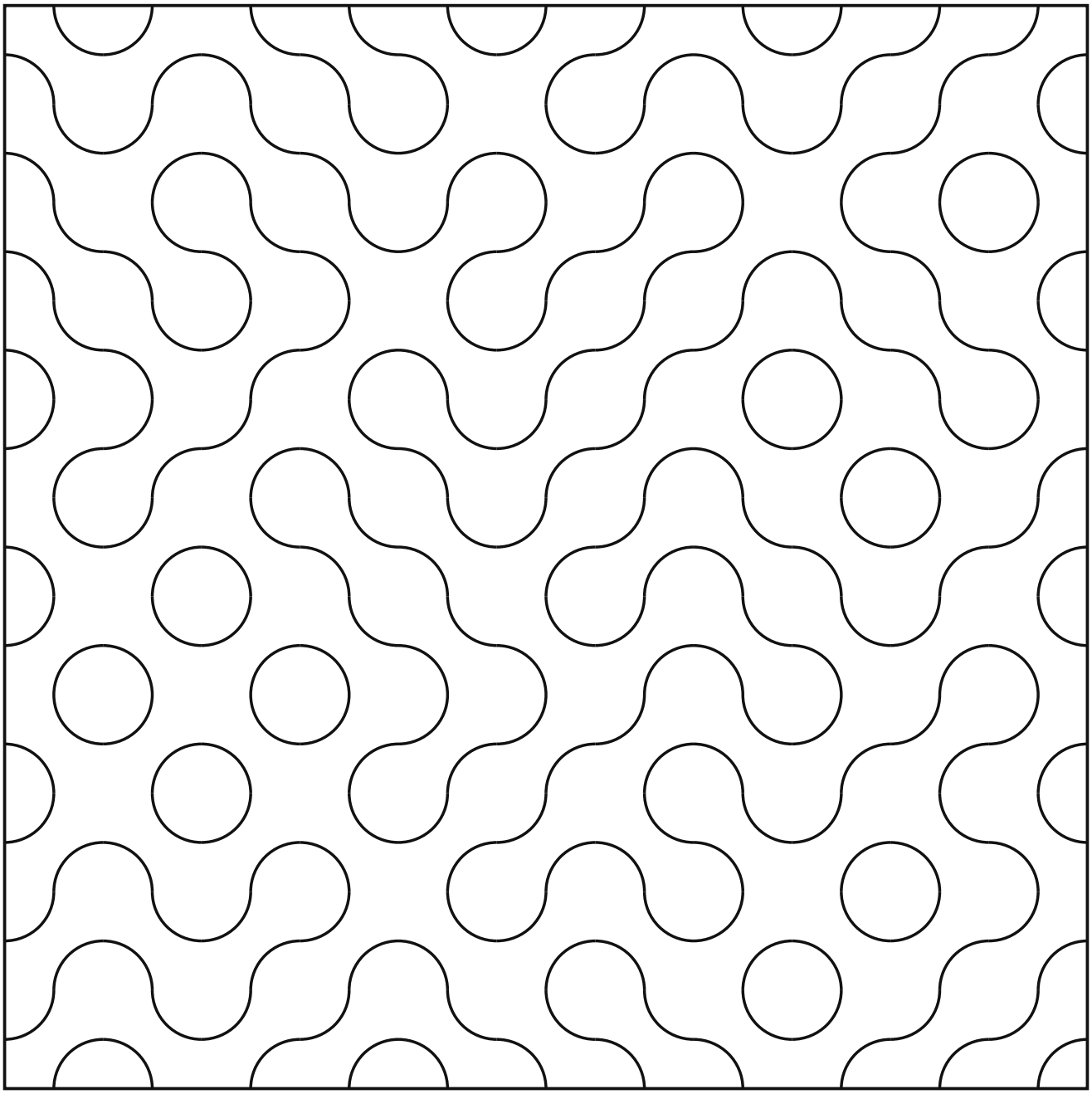
Windswept Tree

Unequal angles make the same rule grow sideways.



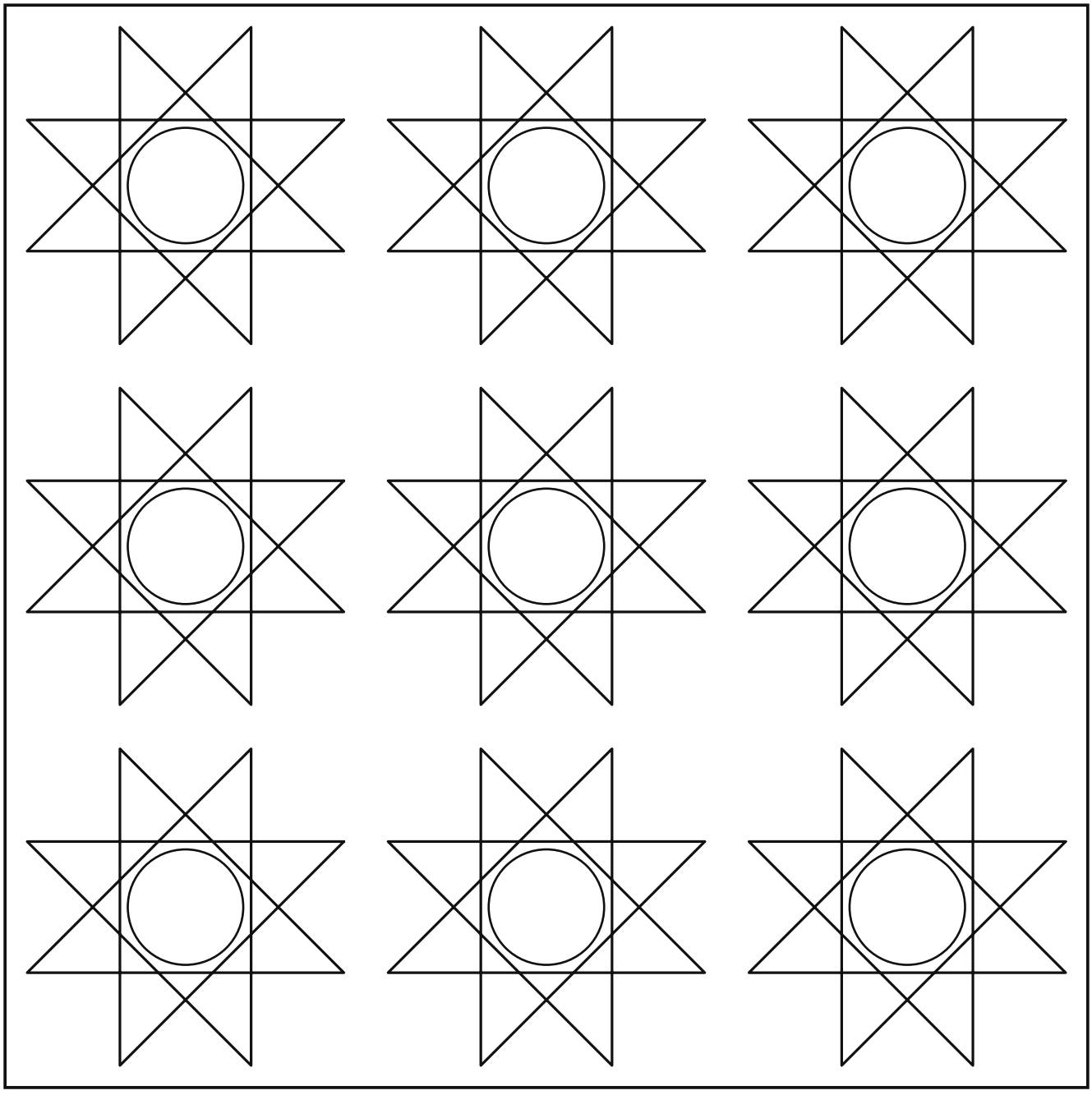
Truchet Tiles I

One square tile, two orientations, endless corridors.



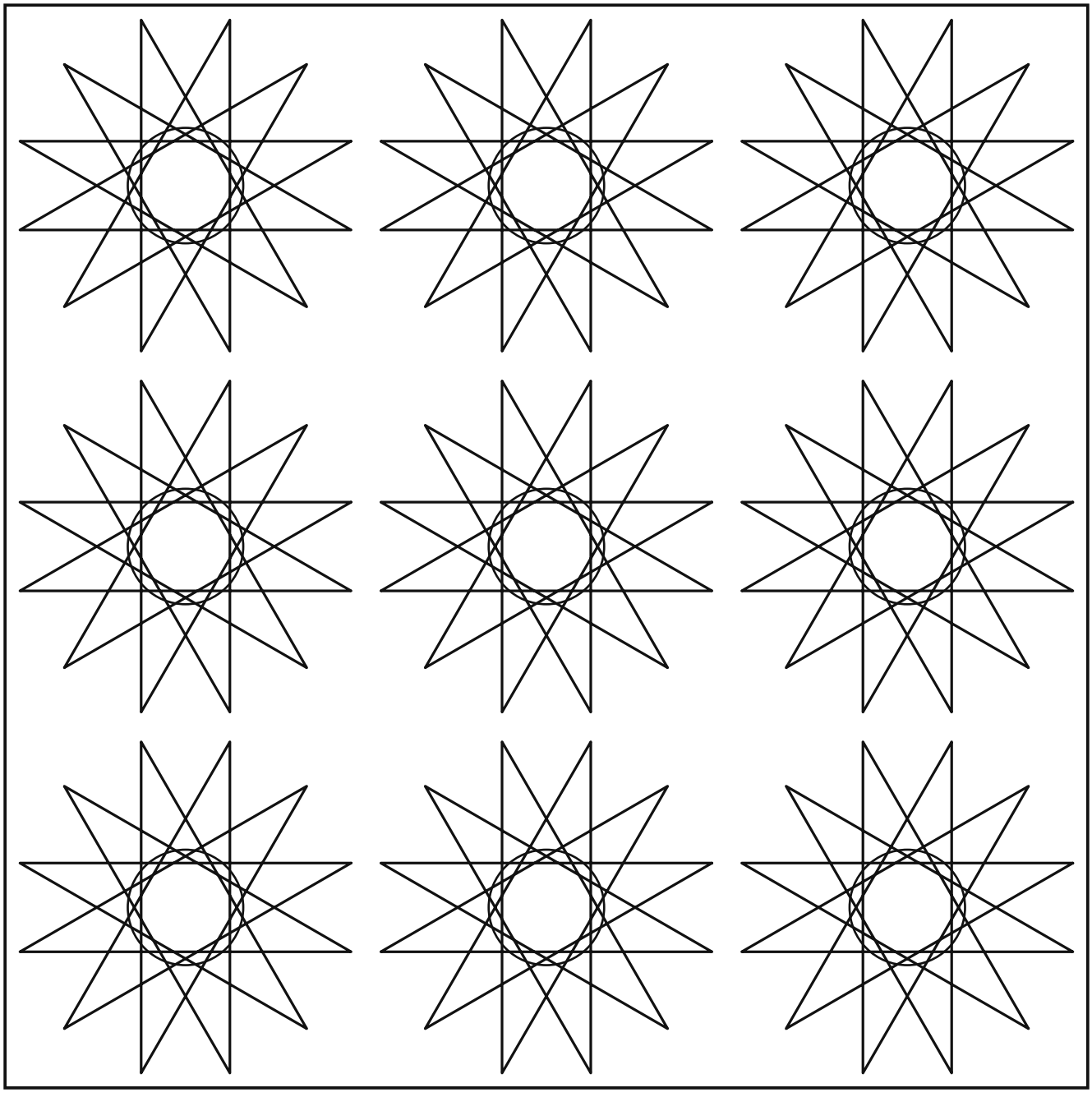
Truchet Tiles II

A different coin-flip sequence, a different maze.



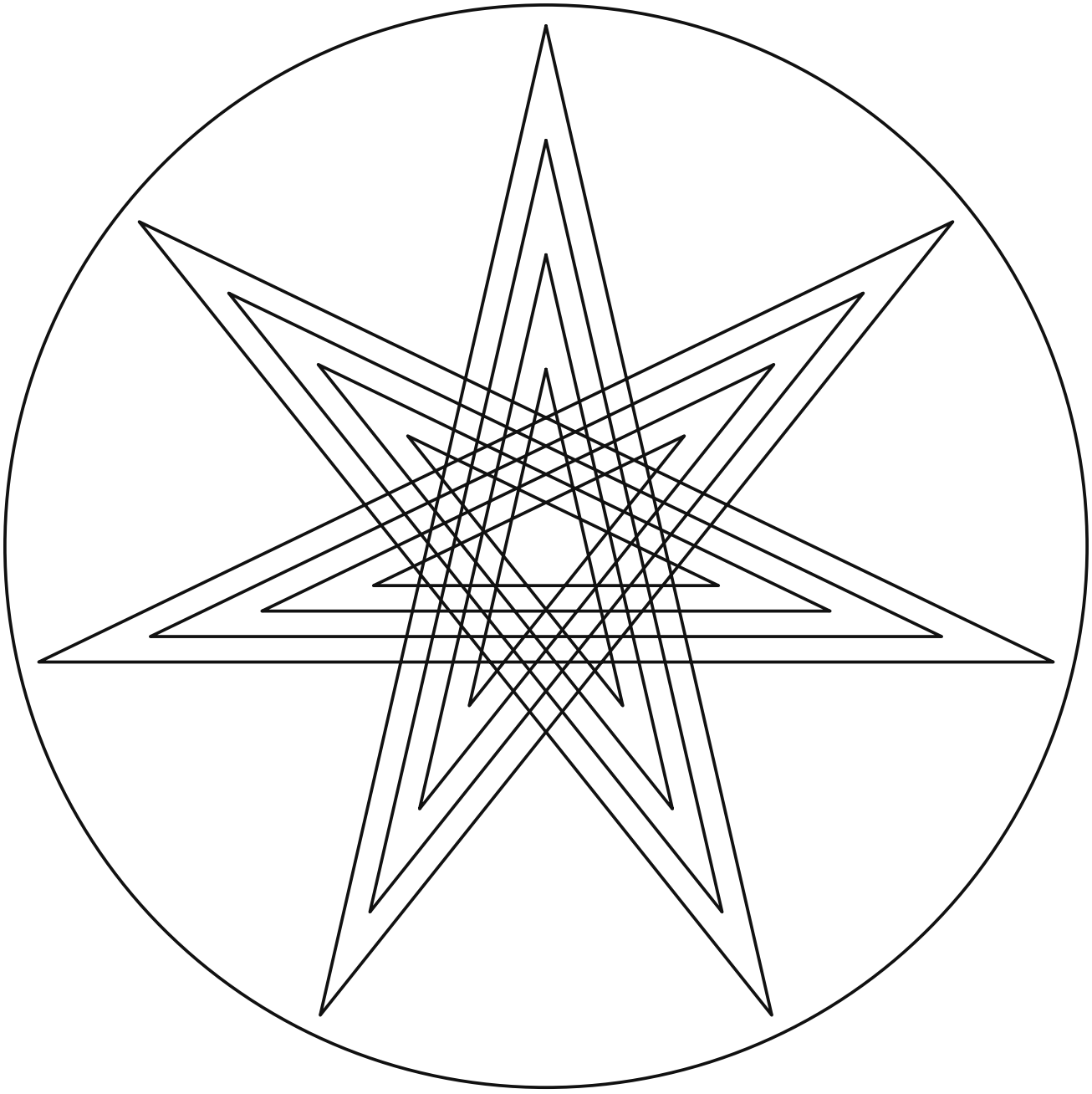
Eight-Point Star Grid

The {8/3} star of Islamic geometric art, repeated.



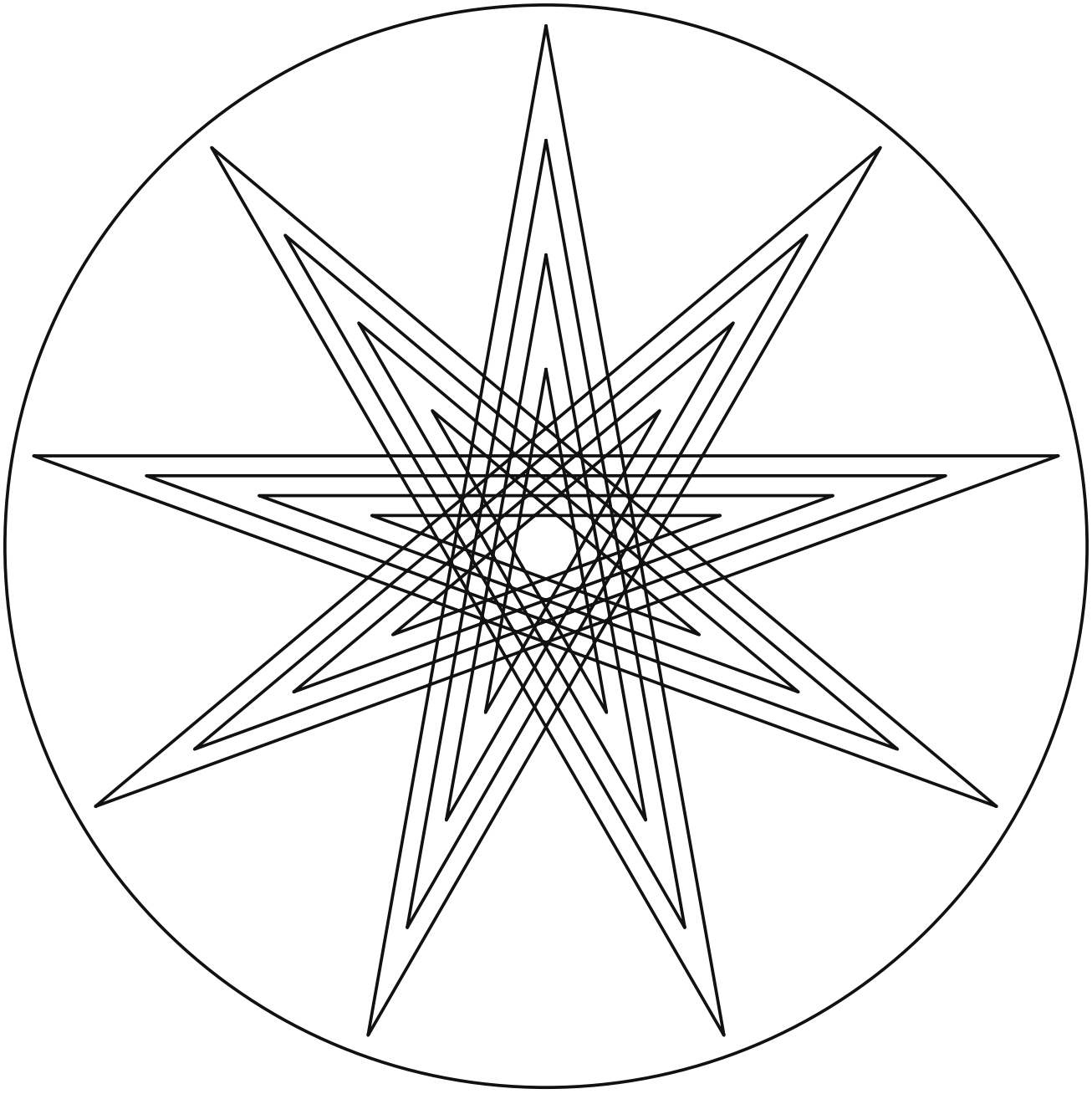
Twelve-Point Star Grid

The {12/5} star — drawn with one unbroken line.



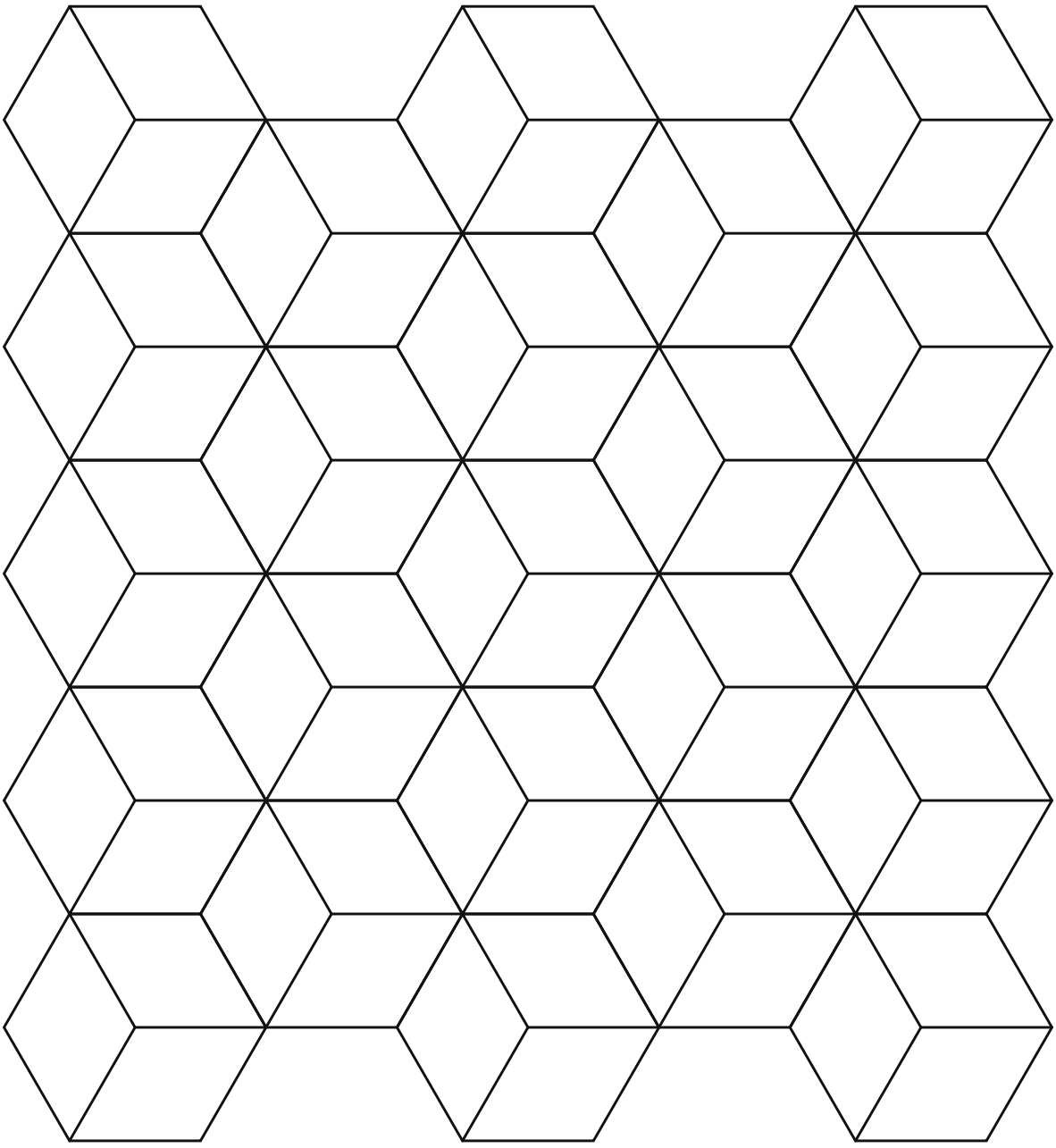
Nested Heptagram

{7/3}: connect every third corner of a heptagon.



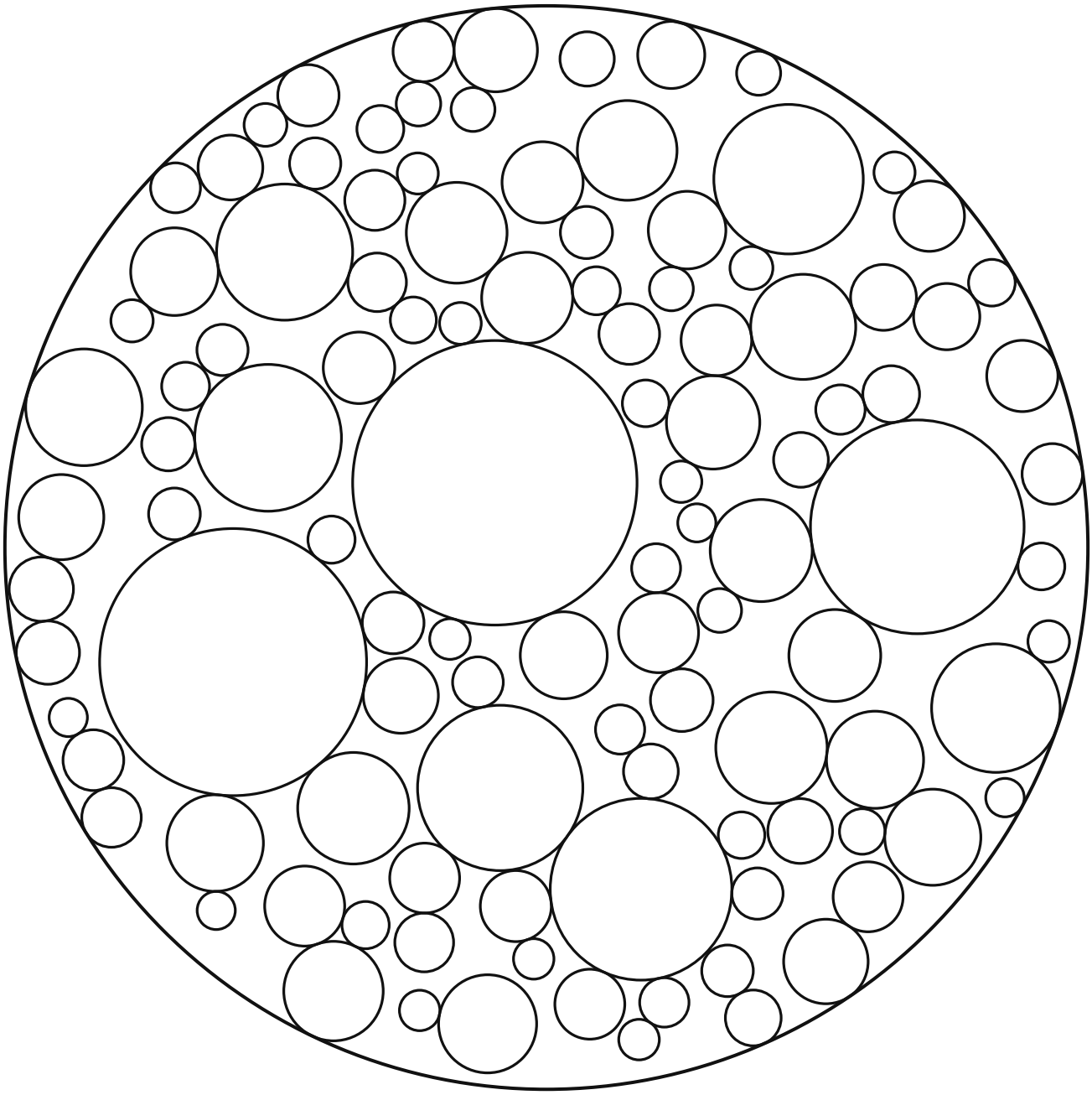
Nested Nonagram

{9/4}: nine points, skip four, never lift the pen.



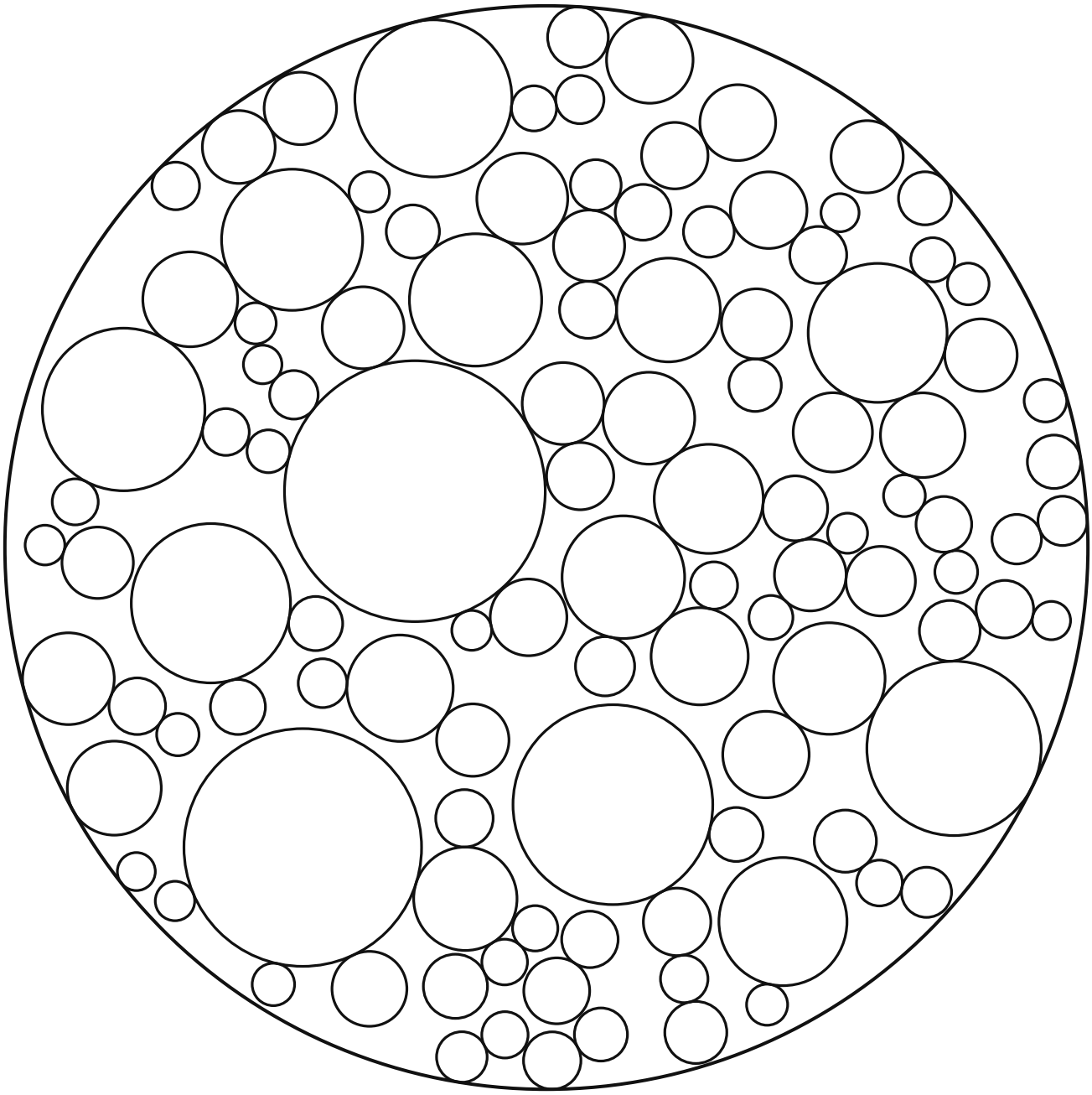
Rhombille Tiling

Three diamonds per hexagon — the tumbling-cubes illusion.



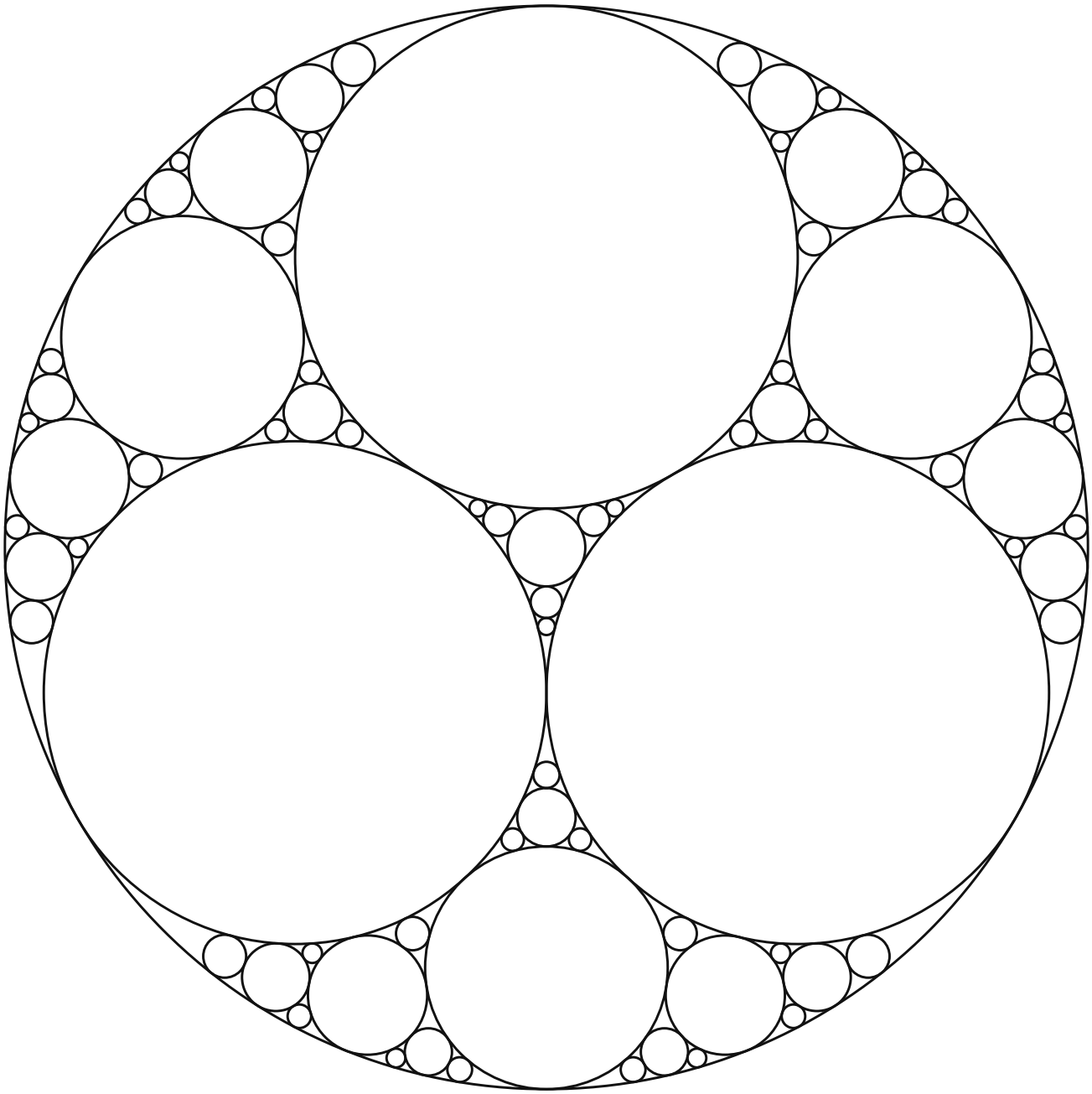
Circle Packing I

Circles grown until they touch — nature's packing.



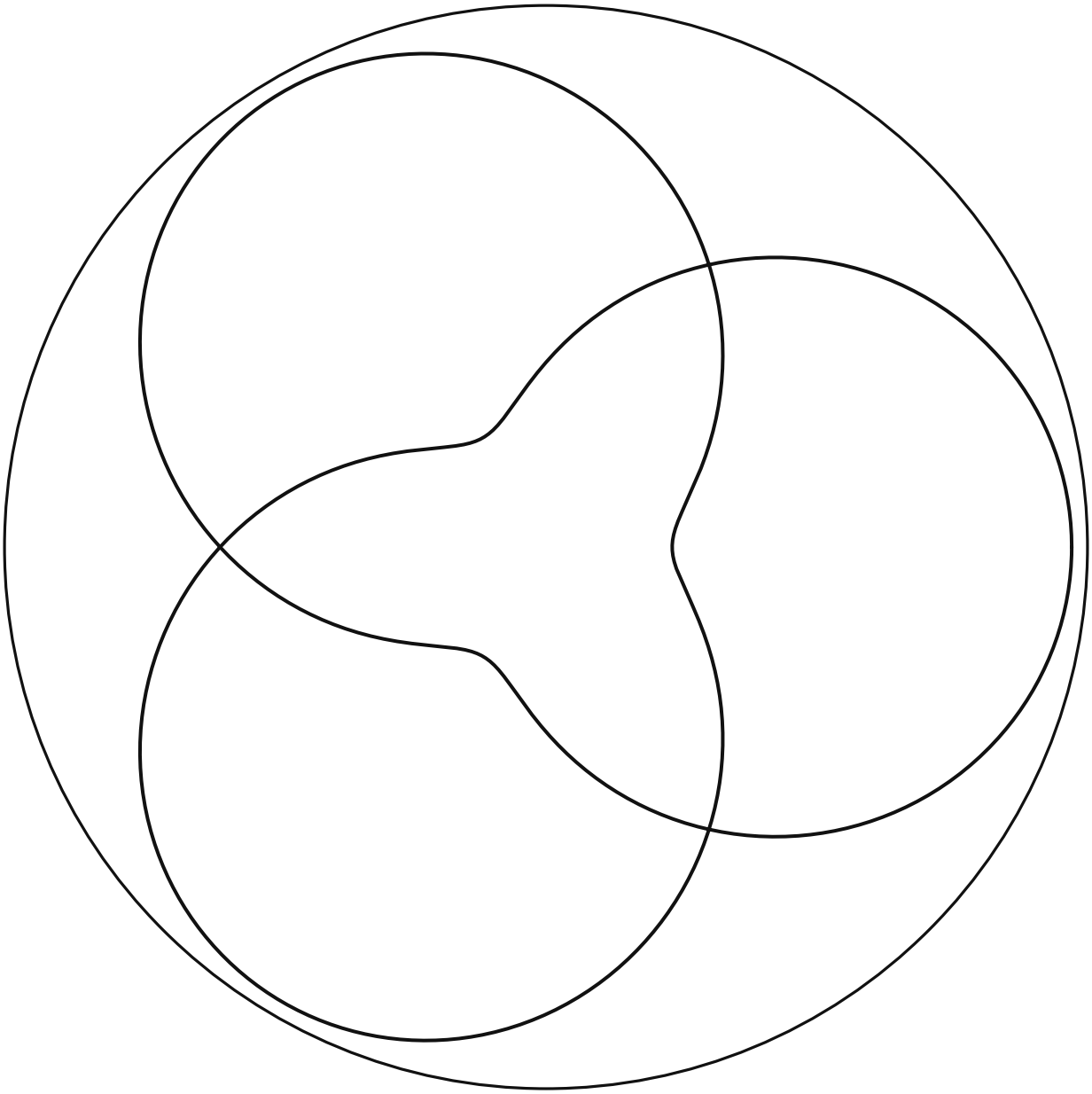
Circle Packing II

A new random seed, a new arrangement of tangencies.



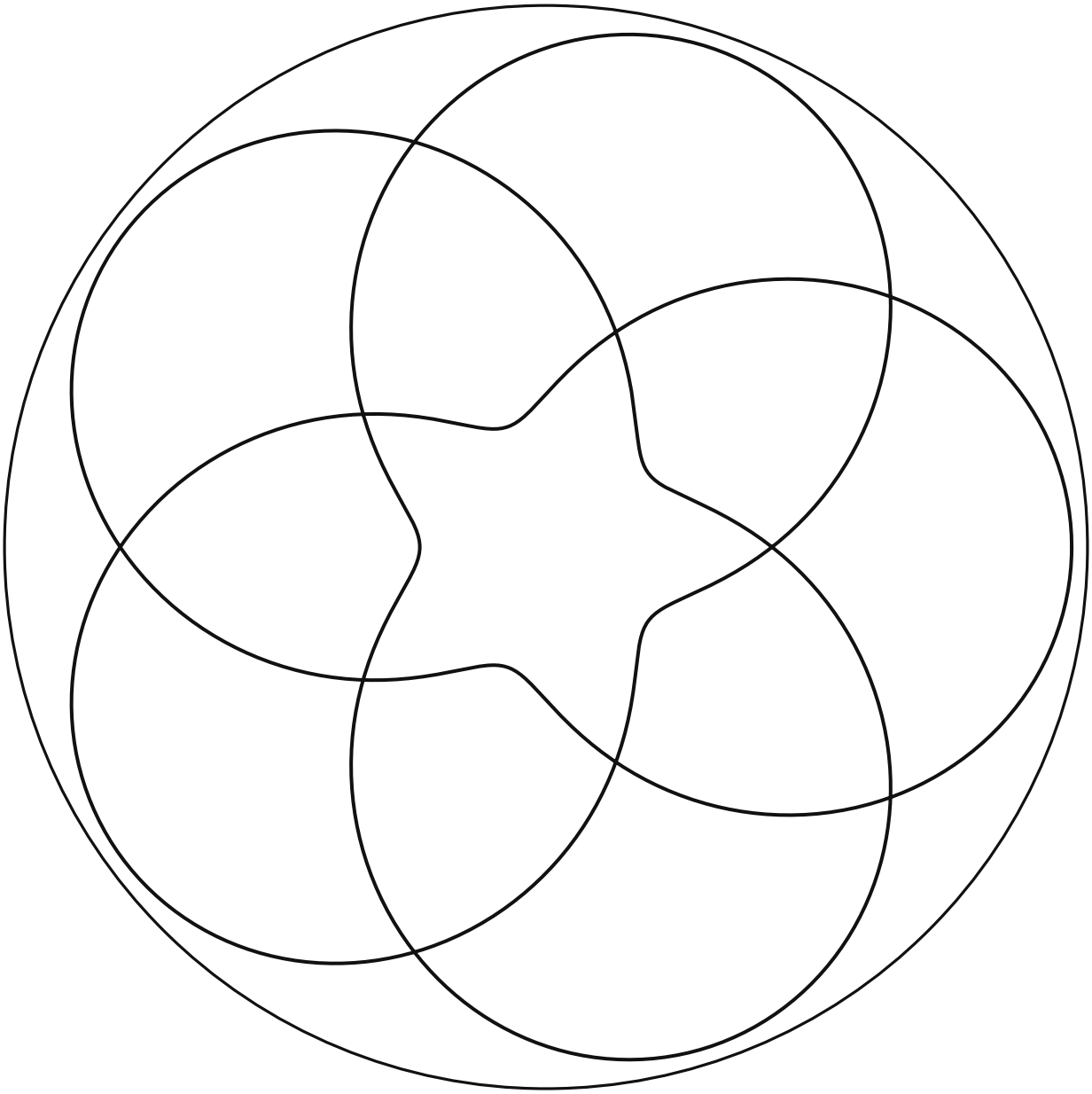
Apollonian Gasket

Every gap between three circles holds another circle.



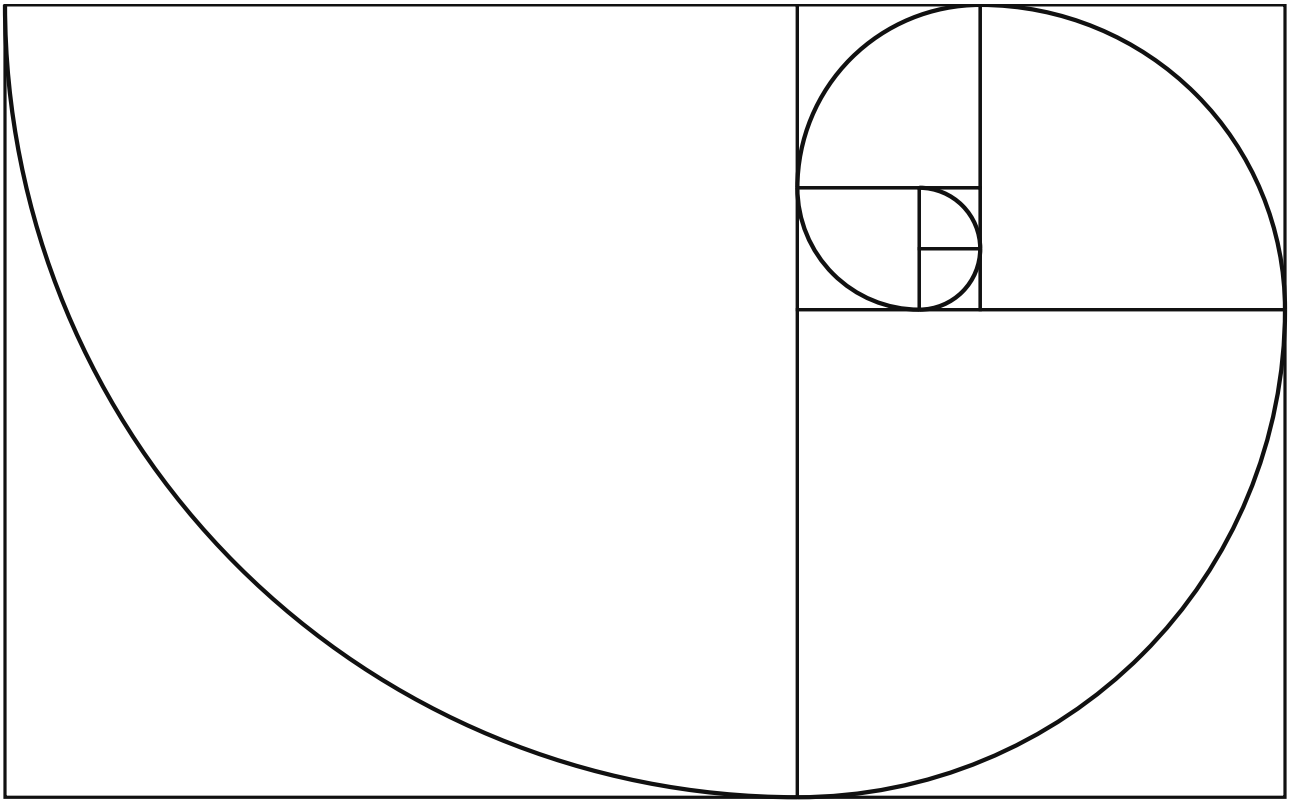
Torus Knot (2,3)

The trefoil: the simplest knot that cannot be undone.



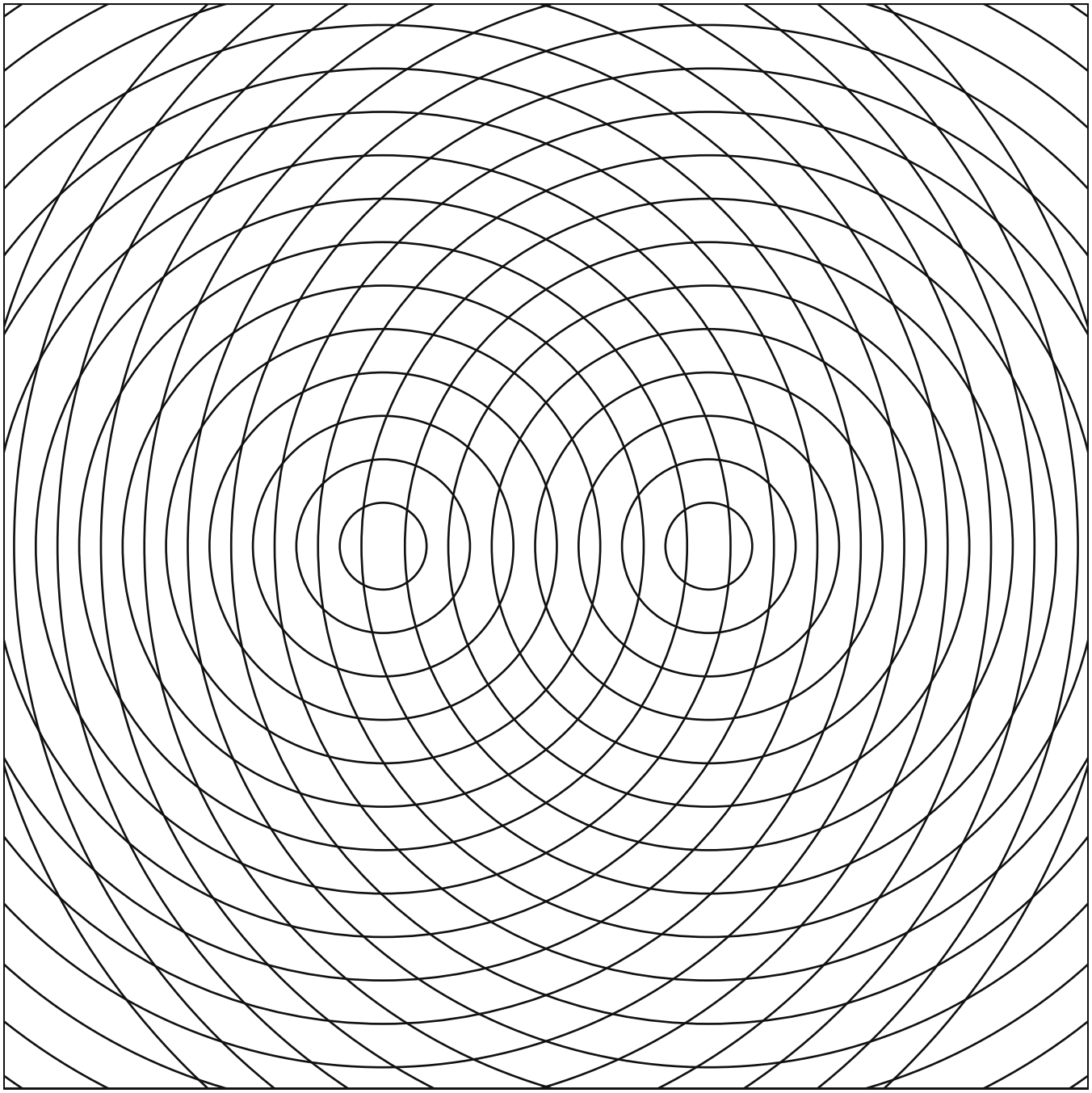
Torus Knot (3,5)

Wind 3 times one way, 5 the other, around a donut.



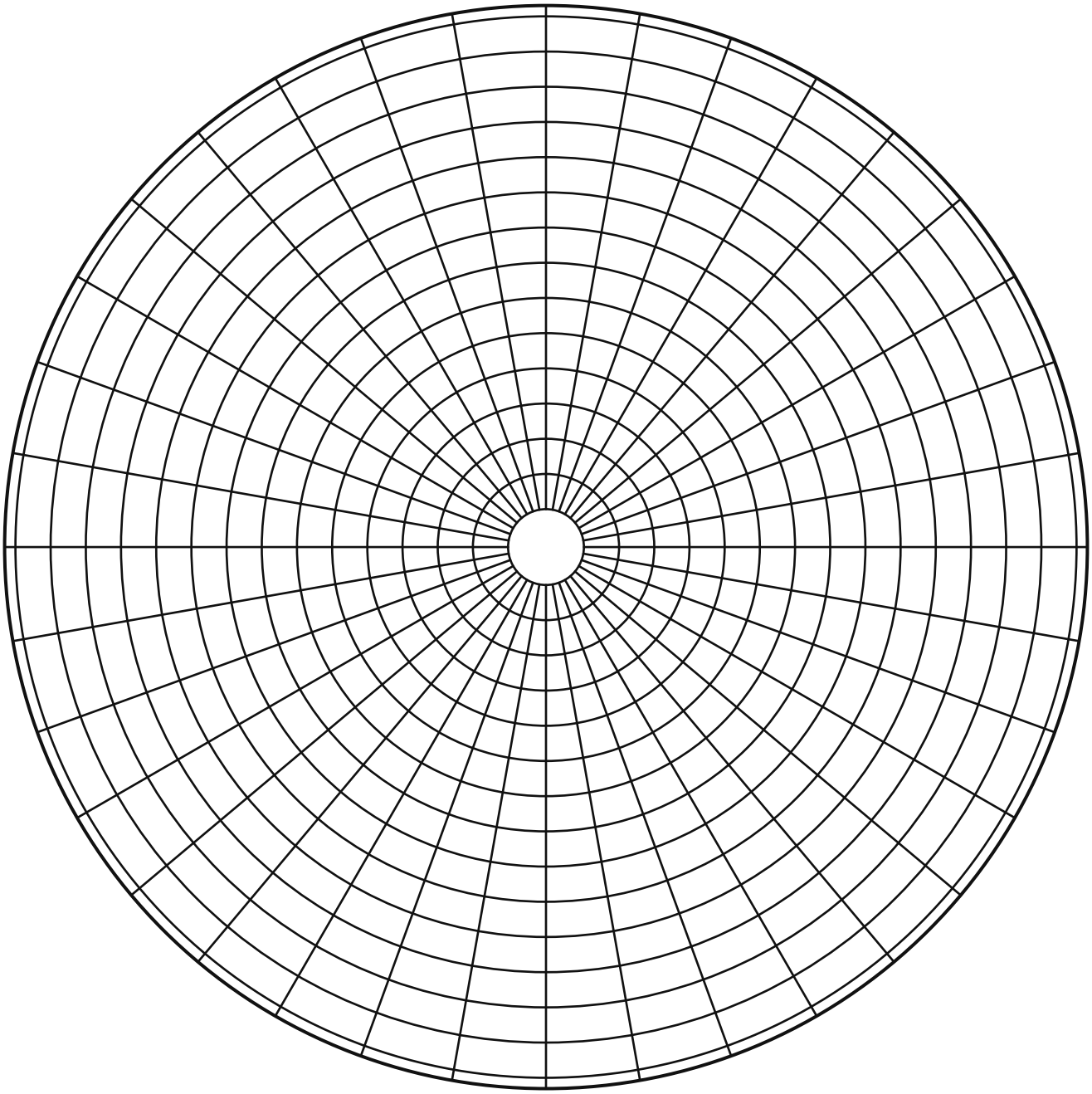
Golden Spiral

Fibonacci squares 1,1,2,3,5,8,13 — the spiral lives in their corners.



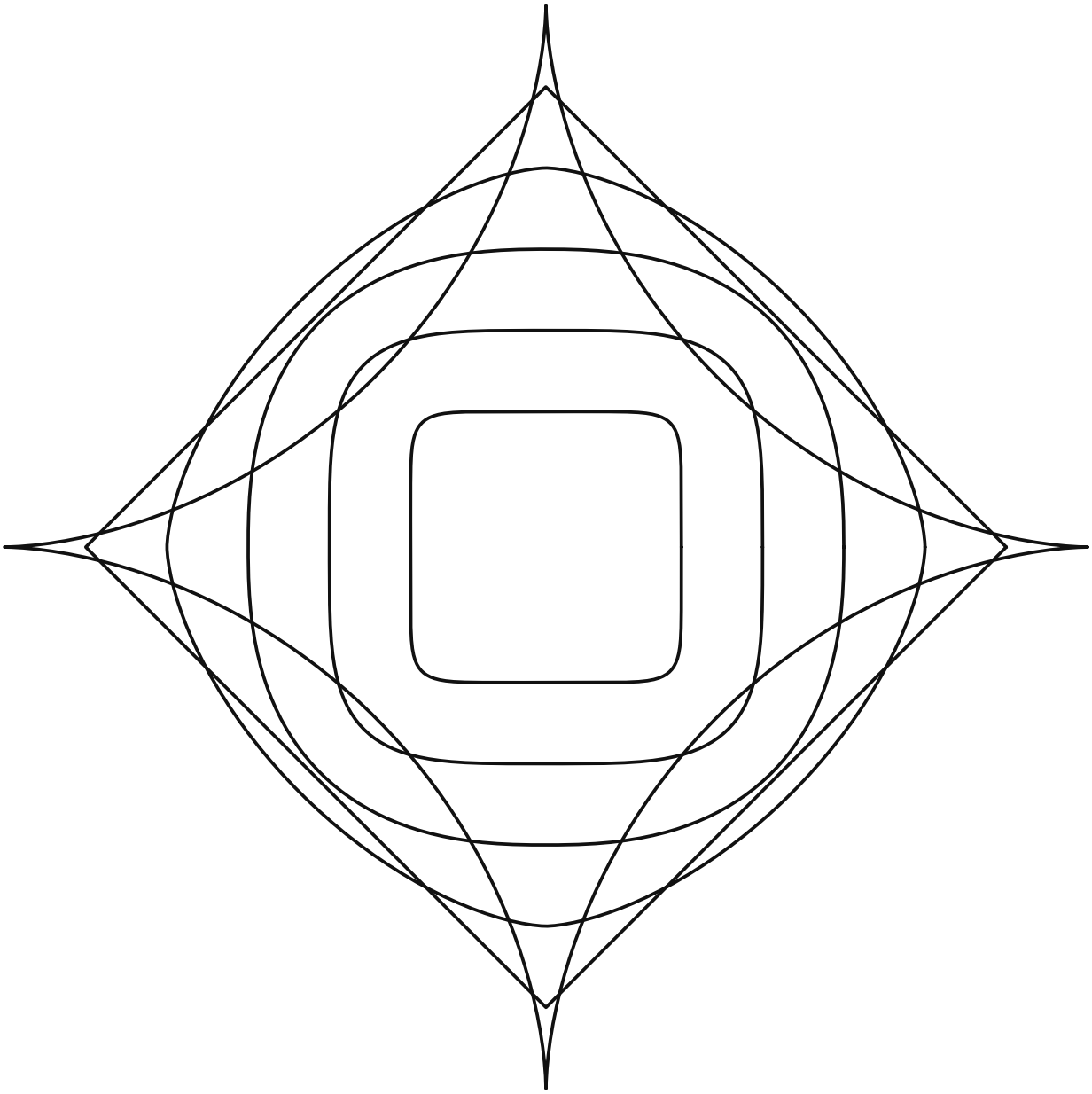
Moiré Circles

Two identical ripple patterns, slightly offset, interfere.



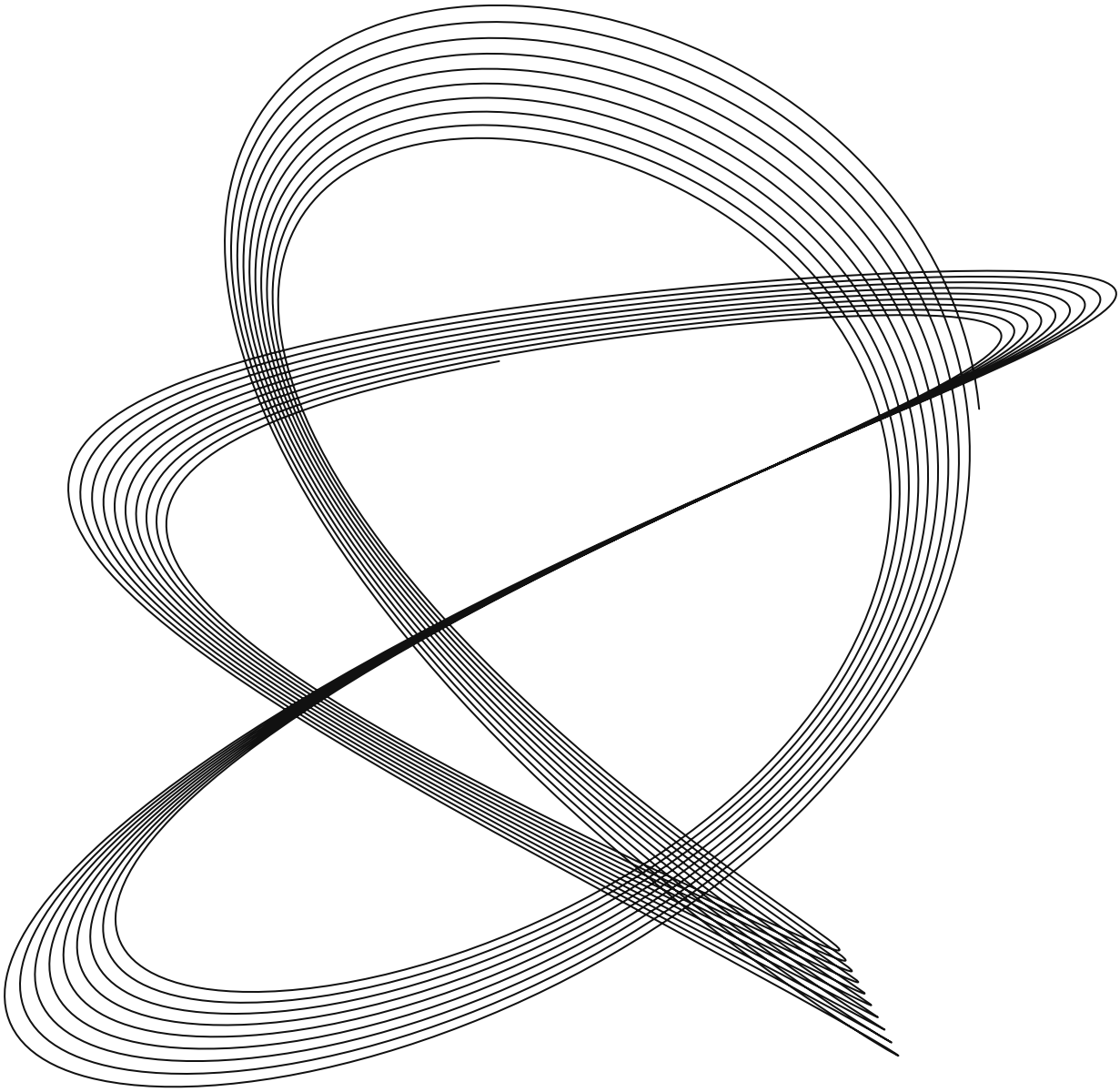
Polar Grid

Circles times rays: the graph paper of polar coordinates.



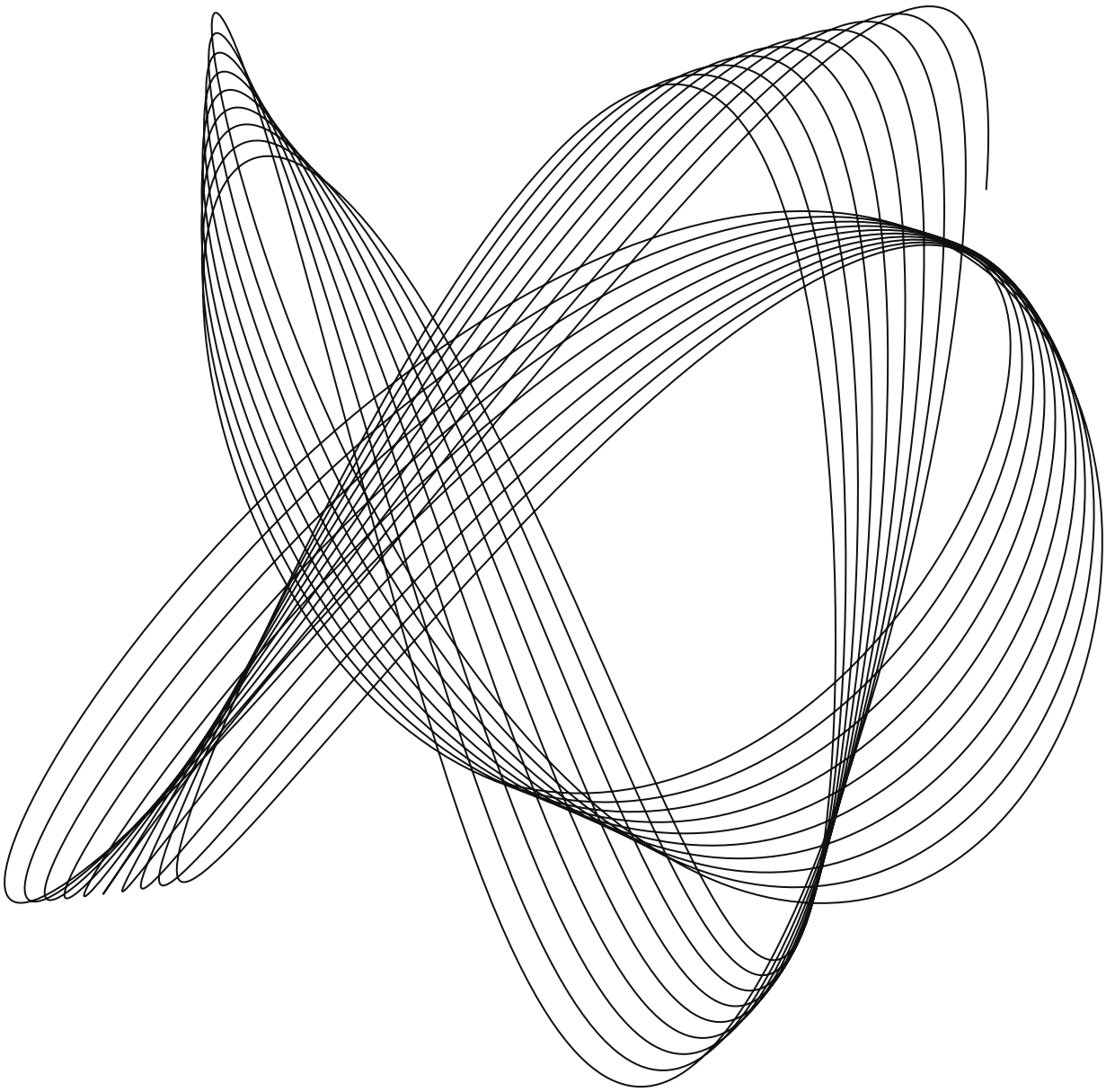
Superellipses

$|x|^n + |y|^n = 1$ morphs a diamond into a square.



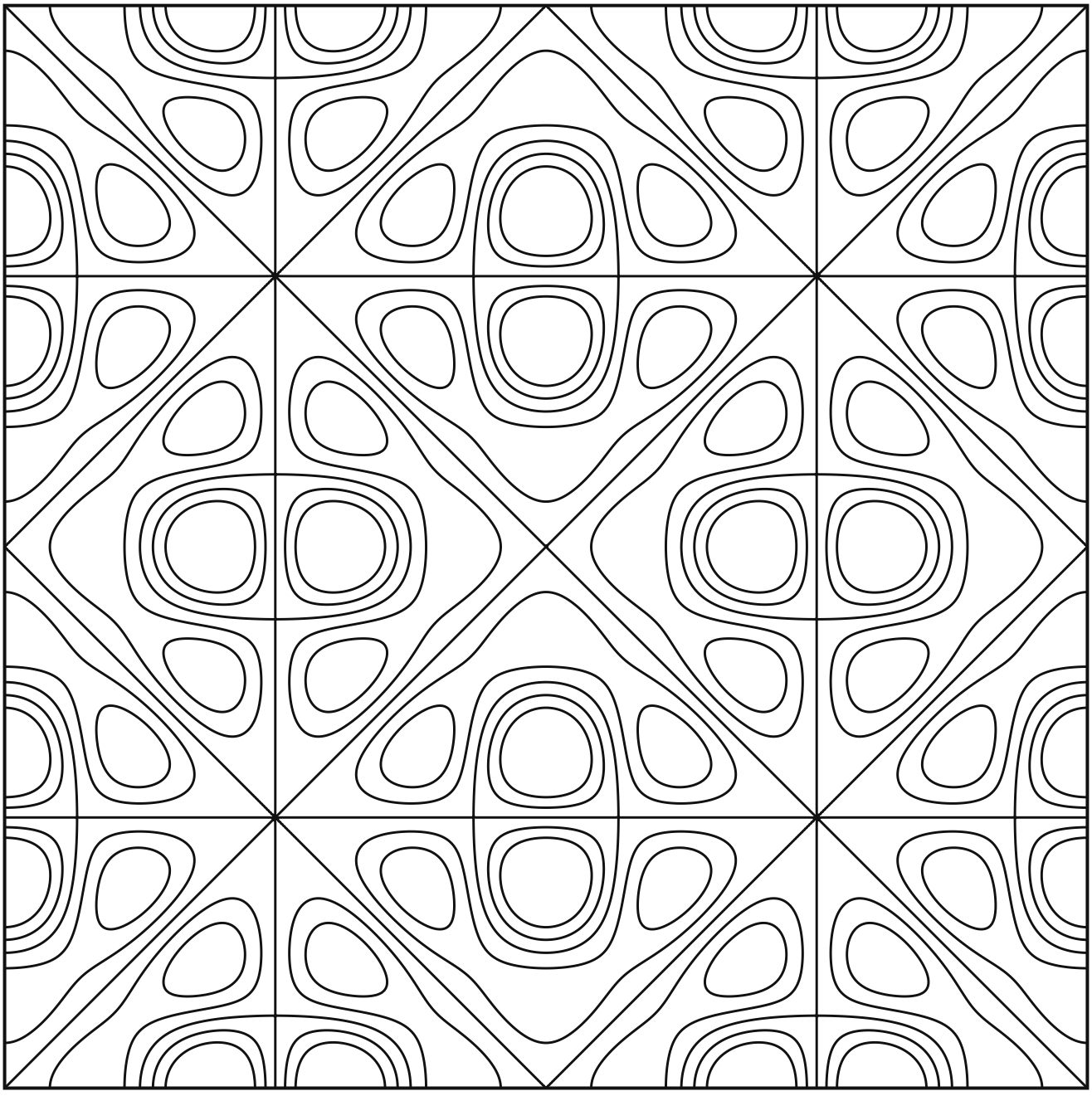
Harmonograph I

A pendulum with a pen: damped sine waves in two axes.



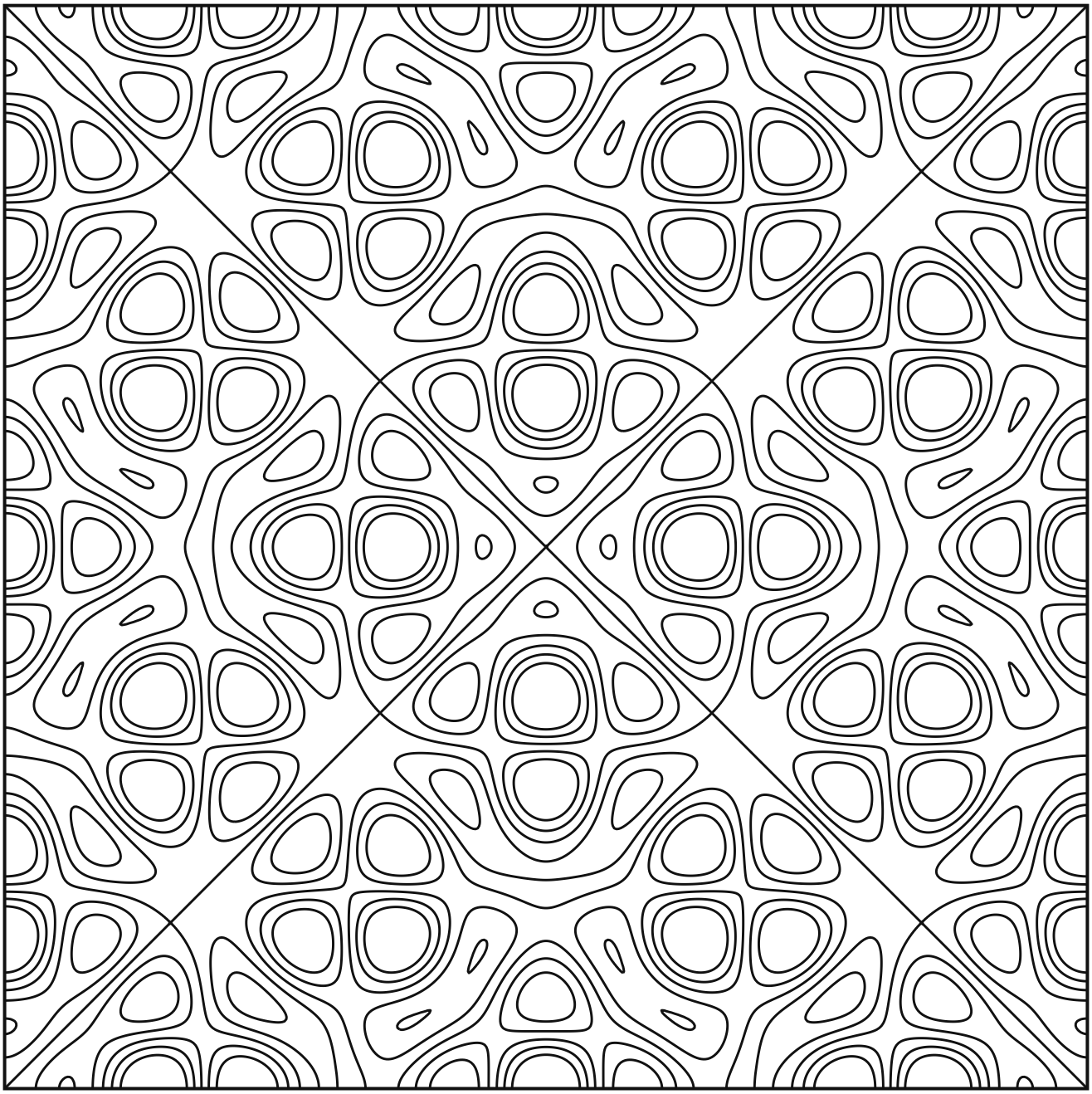
Harmonograph II

Tiny detuning (3.001 vs 2) makes the orbit precess.



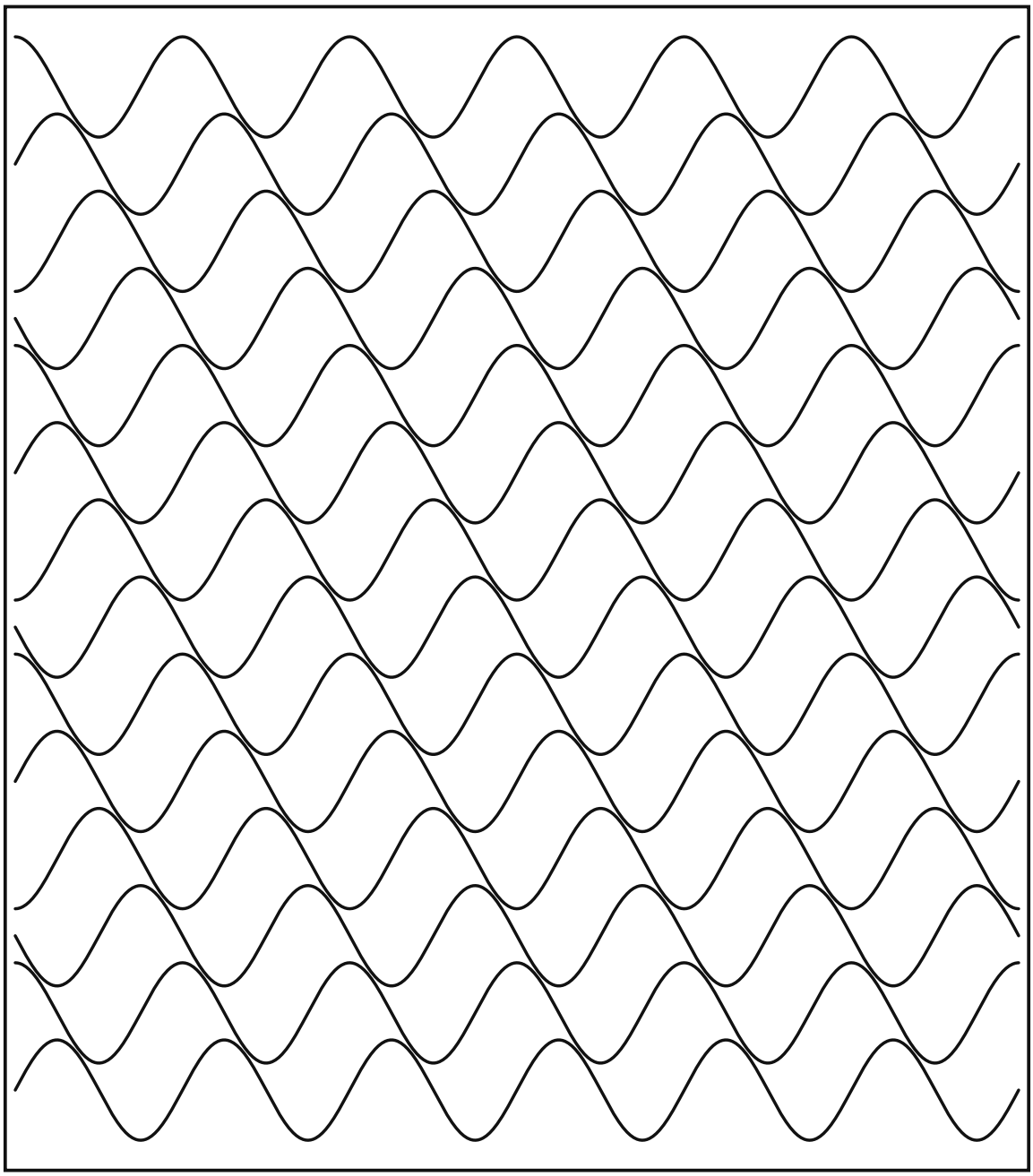
Chladni Figure 3:5

Where a vibrating plate stands still, sand gathers.



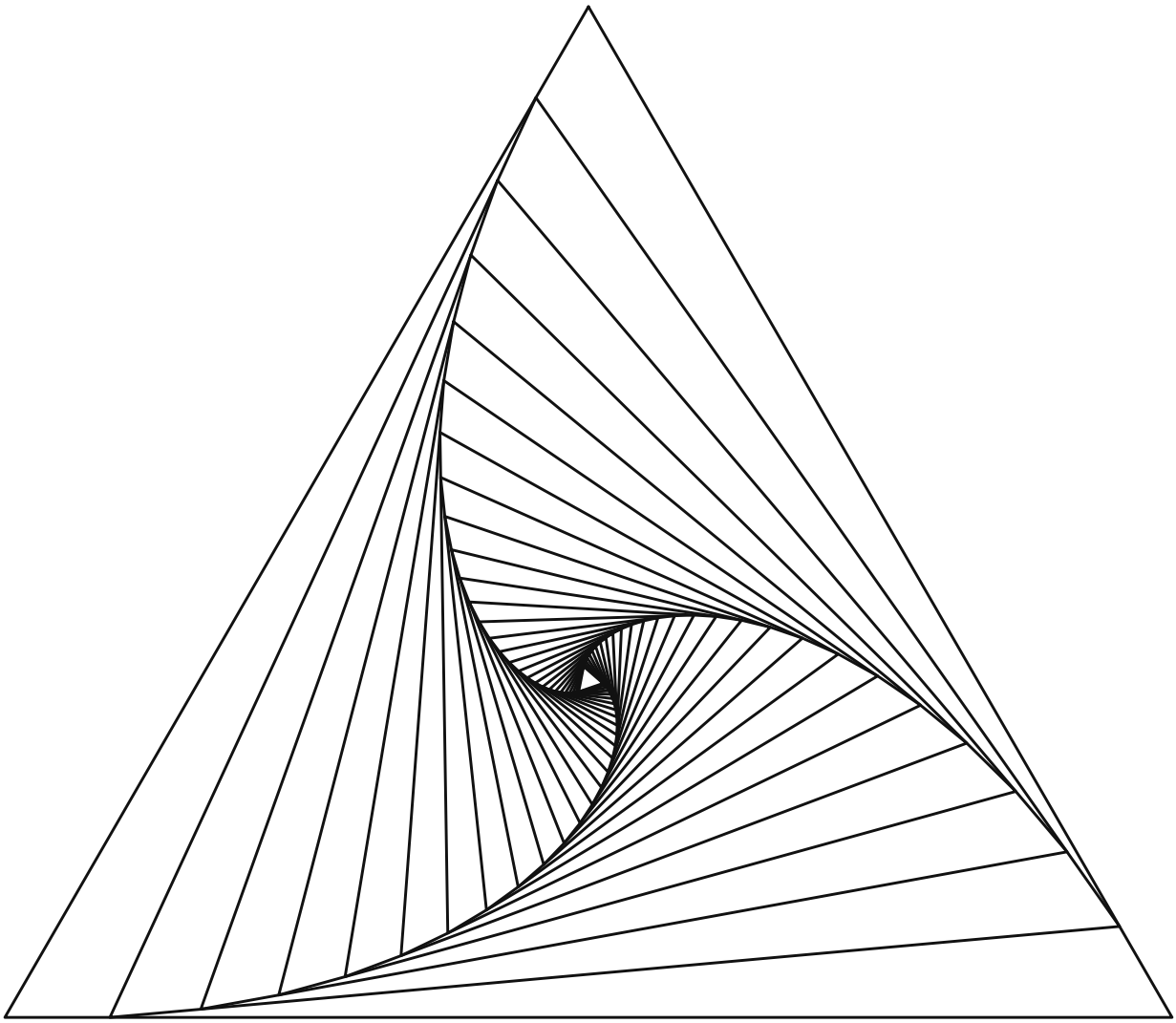
Chladni Figure 4:7

Higher frequencies carve finer standing-wave cells.



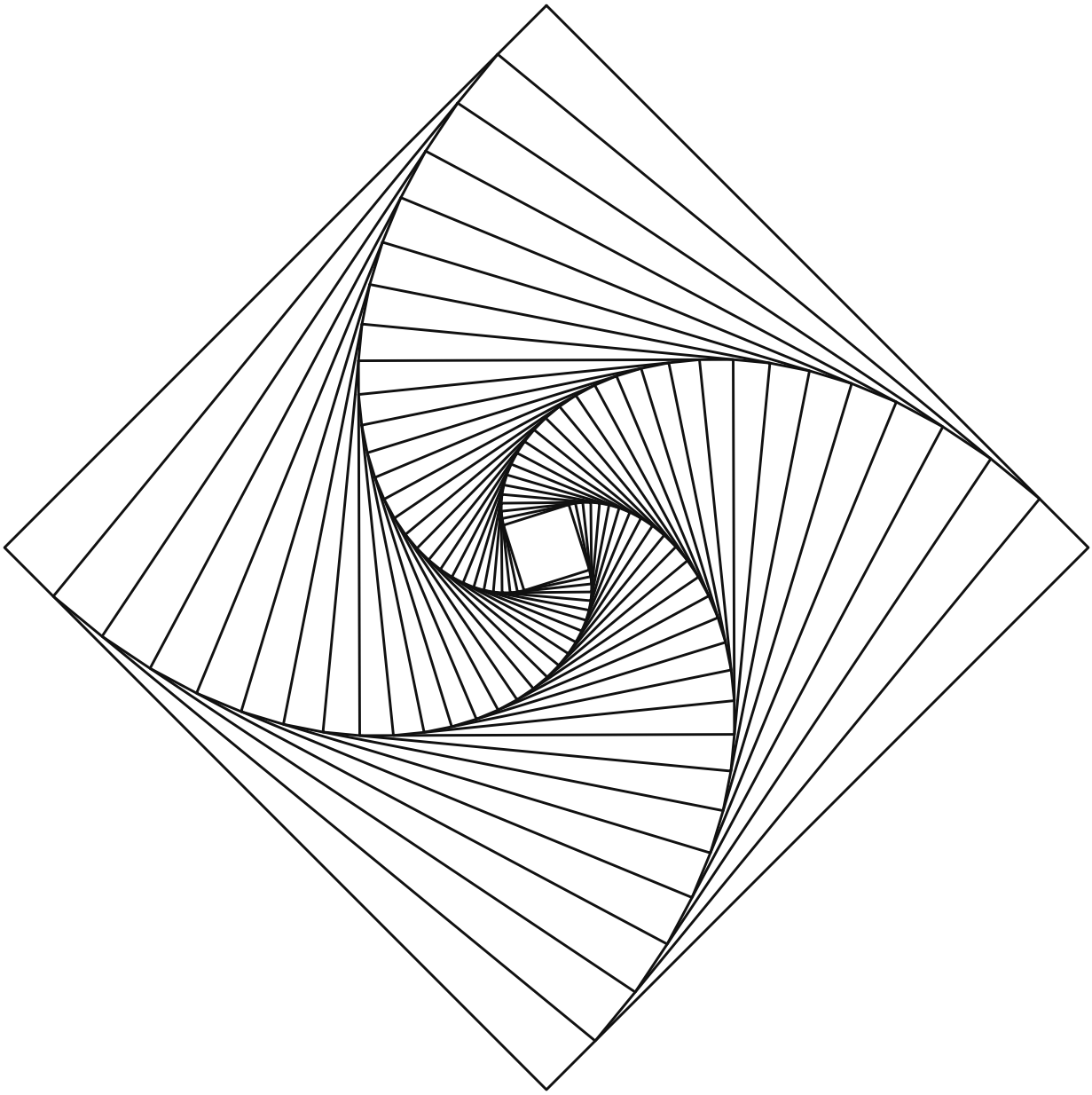
Sine Weave

Fourteen phase-shifted sine waves braid a fabric.



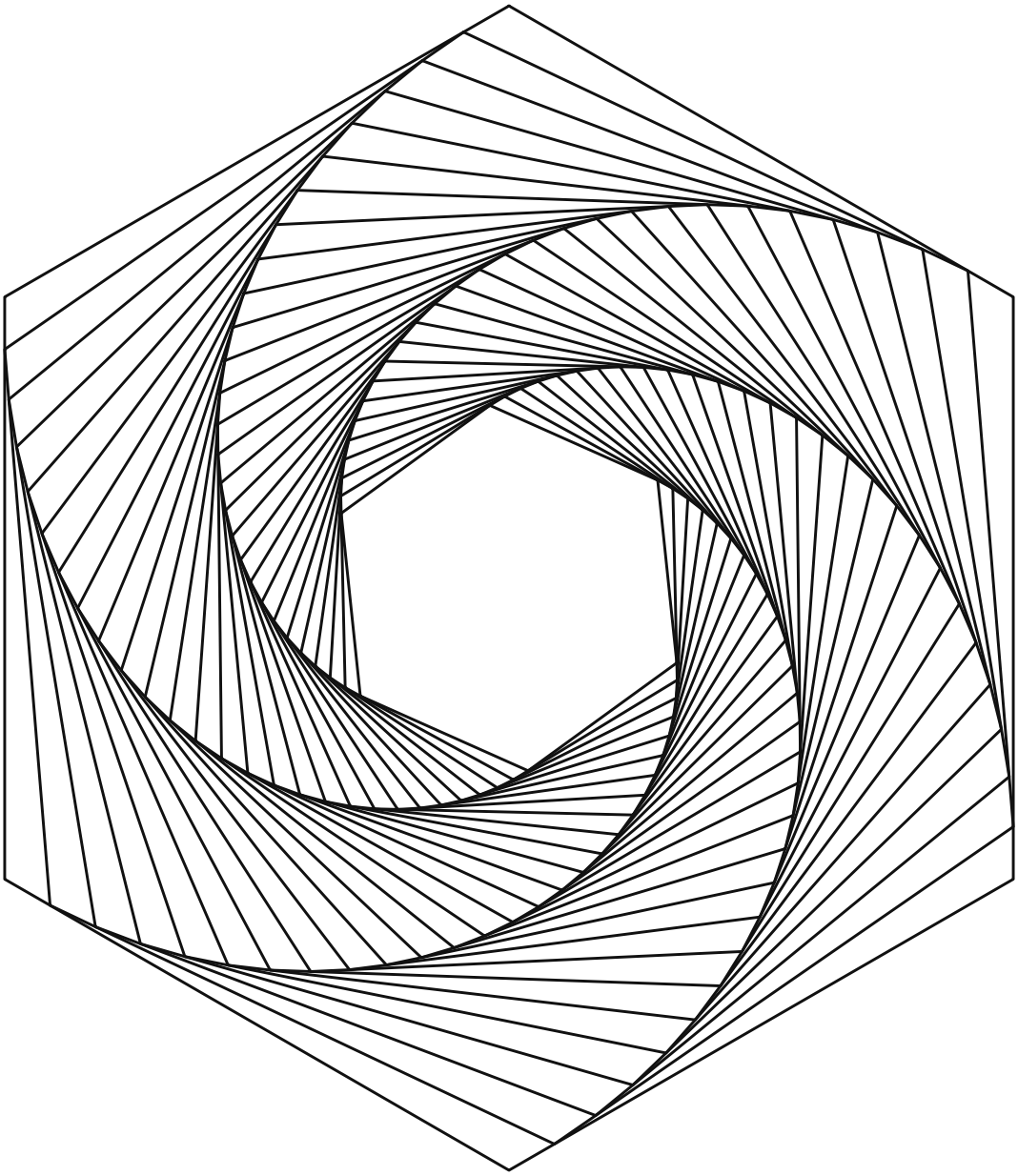
Whirling Triangles

Chase the next corner by 9% each step and spiral in.



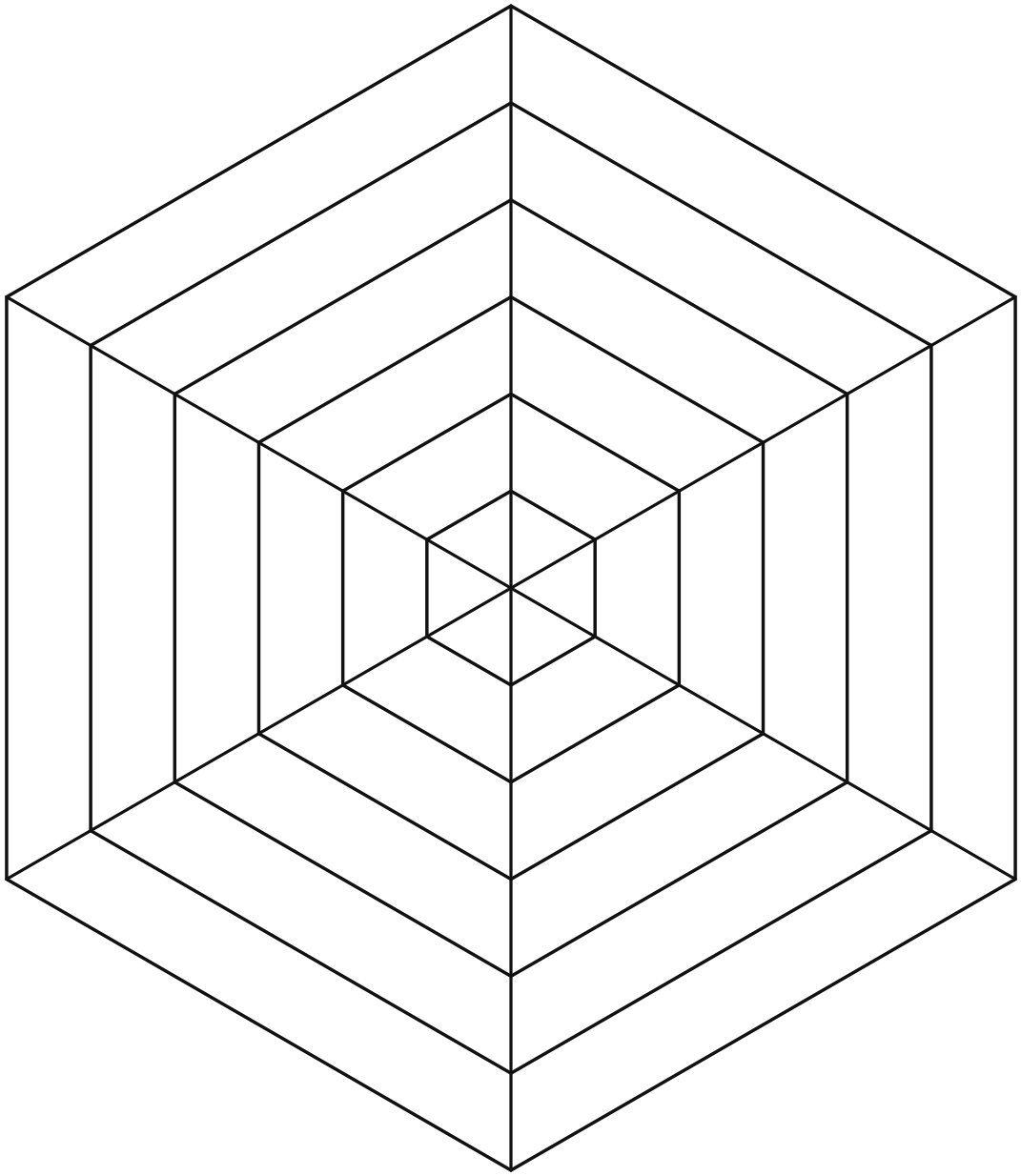
Whirling Squares

The pursuit curve: four mice each chasing the next.



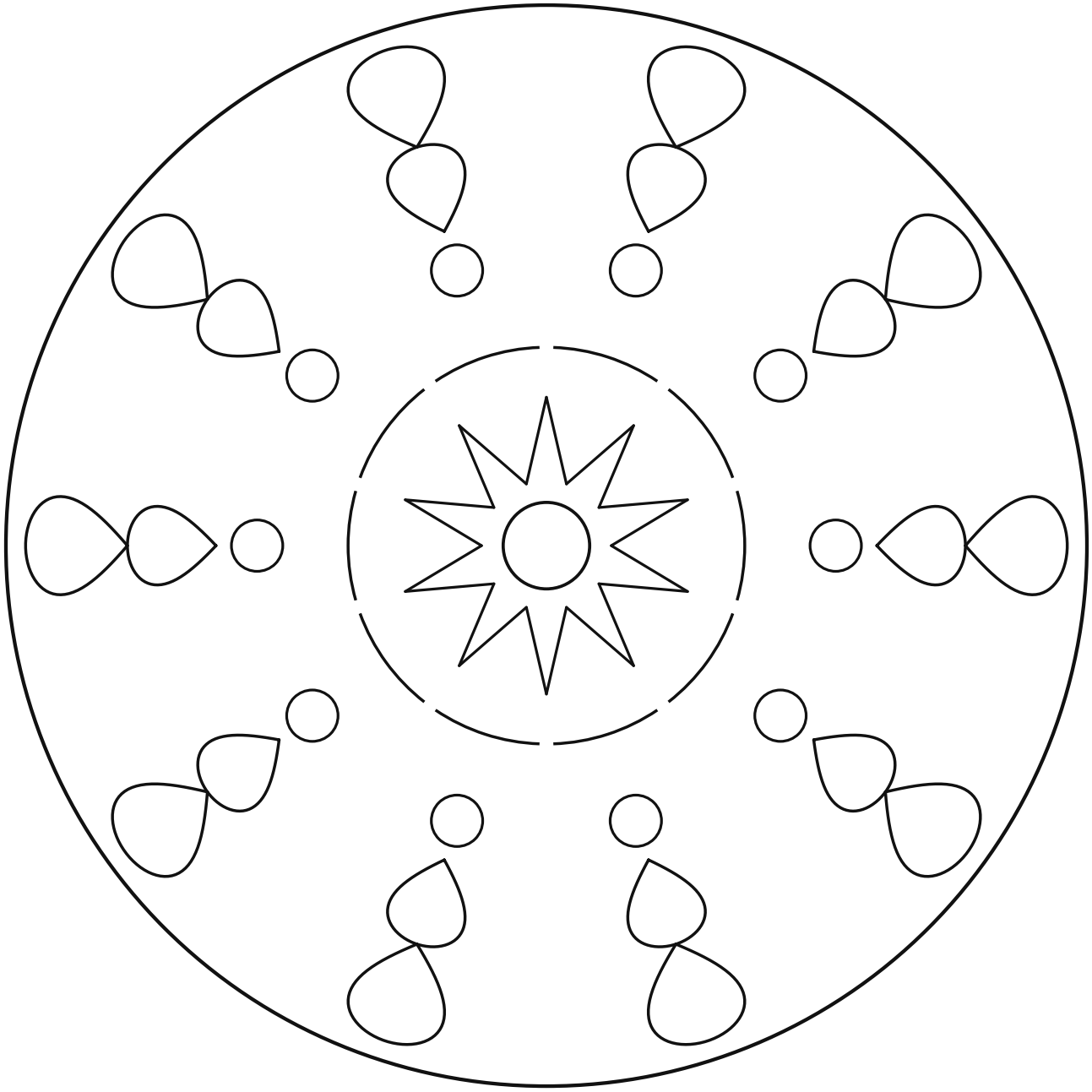
Whirling Hexagons

Six chasers trace a logarithmic spiral nest.



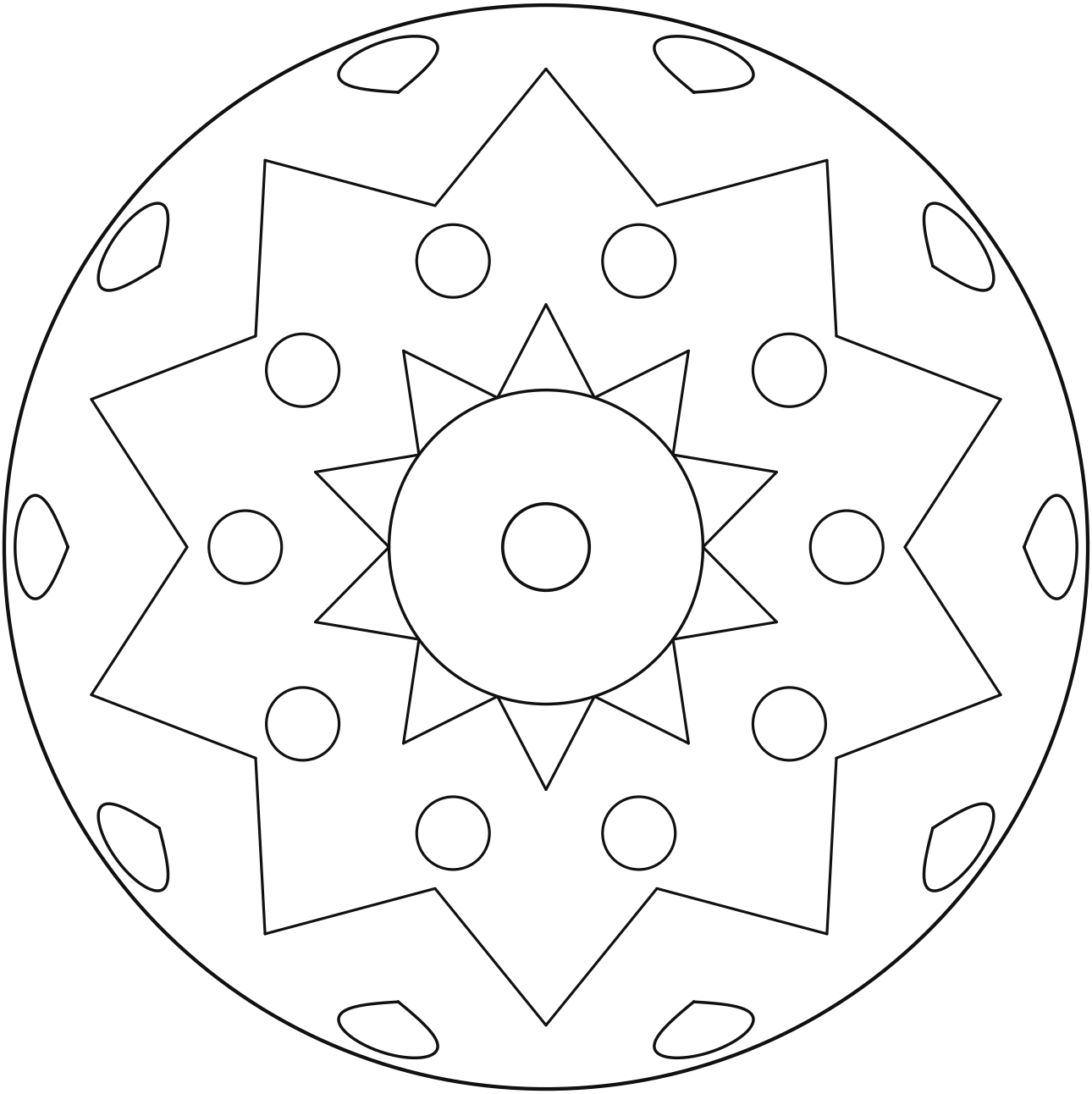
Honeycomb Web

Nested hexagons with spokes — bees build with least wax.



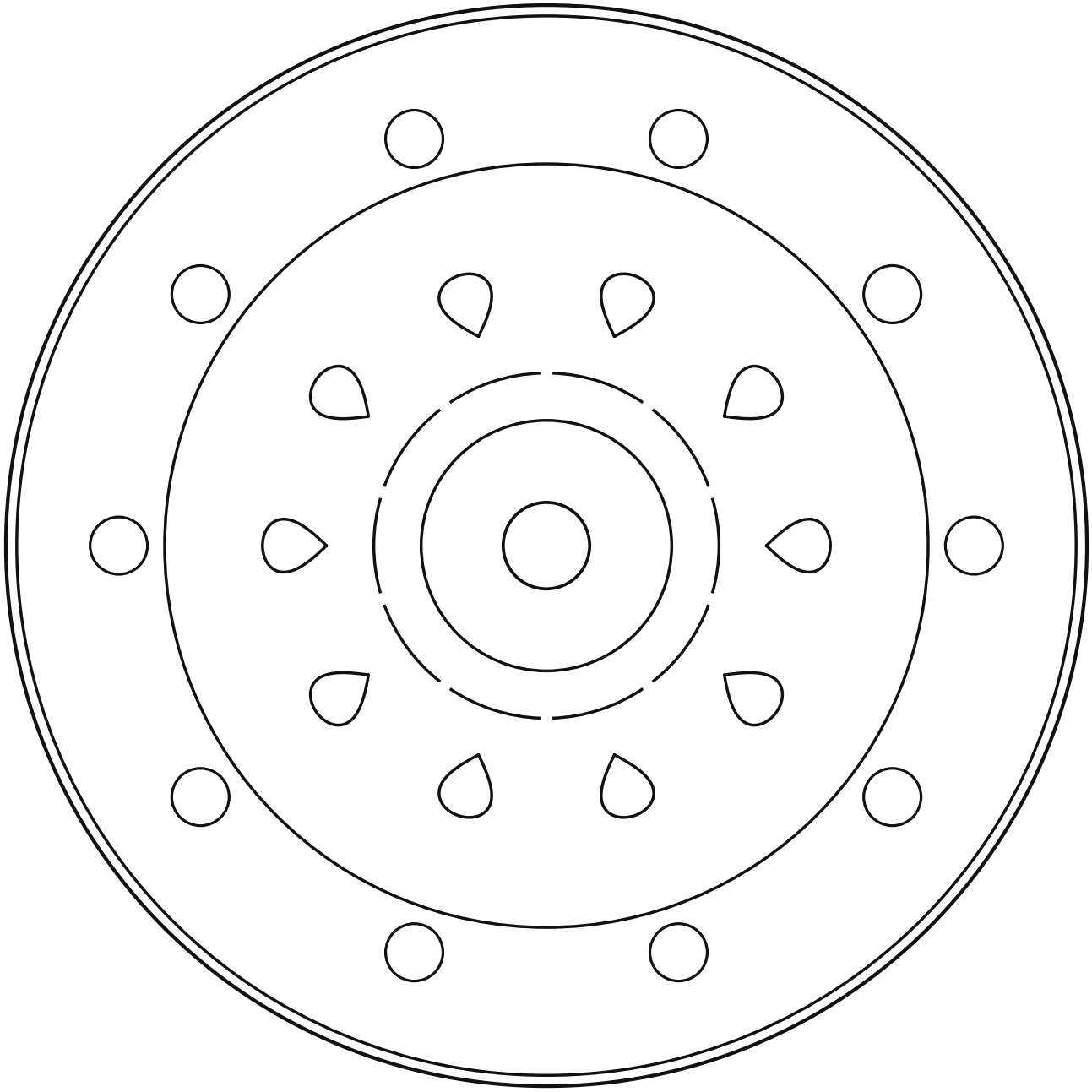
Mandala No. 1

Rotational symmetry: one motif, repeated n times around.



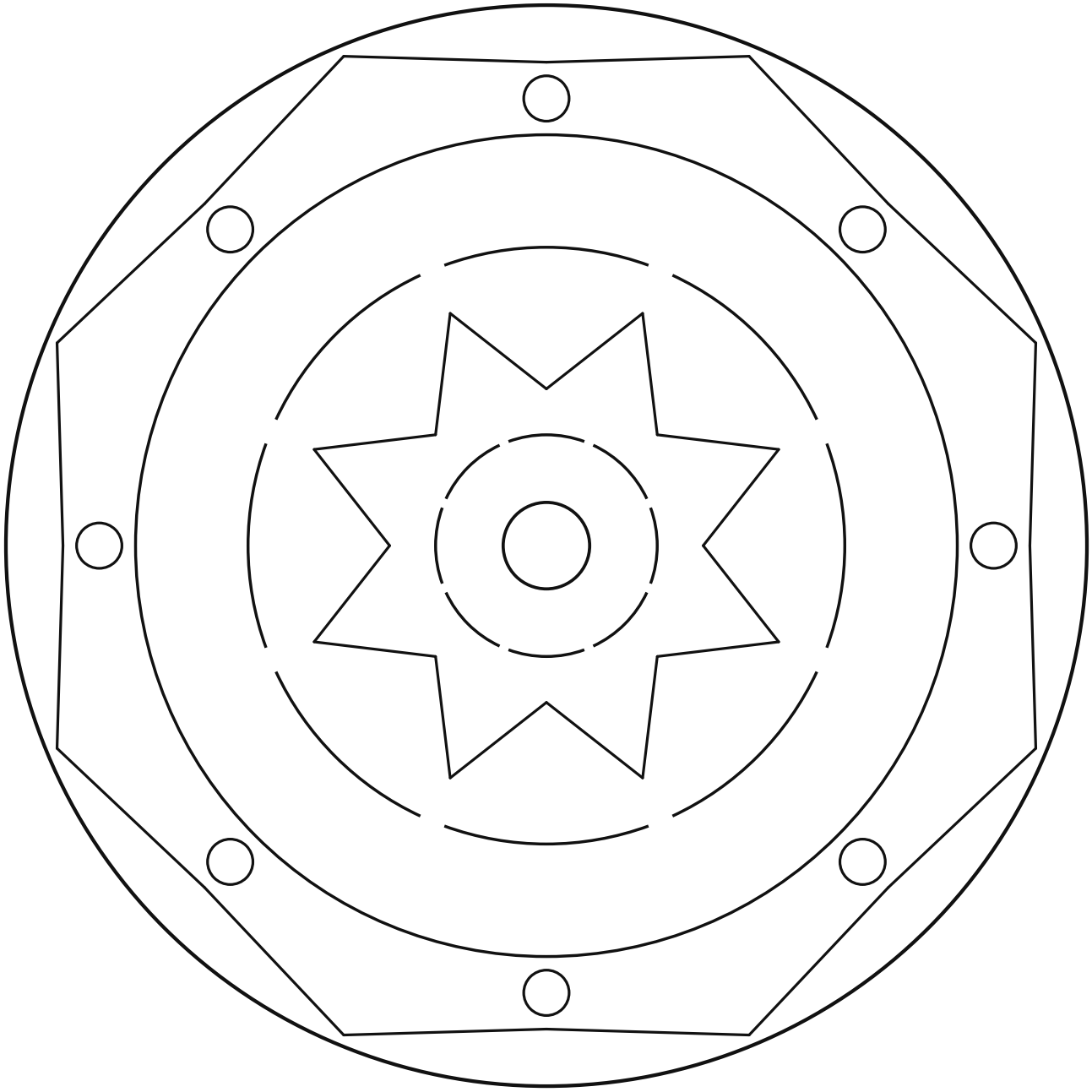
Mandala No. 2

Rotational symmetry: one motif, repeated n times around.



Mandala No. 3

Rotational symmetry: one motif, repeated n times around.



Mandala No. 4

Rotational symmetry: one motif, repeated n times around.